WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZNS-116 (PAD #81), TYPE: IIIF

Drilling Department

JULY 2024

MSA/OEOC			
	Position	**Name**	**Date & Signature**
Prepared by	Drilling Engineering Department	Masoud Babakhani M.H.Haji Taleb	
Approved by	Project Head of Drilling Engineering	Mostafa Khedri	
	Project Head of Drilling Operation	Piran Farzaneh	
	Project Manager	Rashid Ansari	
PPL			
	Position	**Name**	**Date & Signature**
Check & Approved by	Drilling Operation Manager	Majid Nariman	
	Subsurface Engineering Manager	Mahdi Salimmehr	
	Project Manager	Ali Maghzi	
	Subsurface Project Manager	Alireza Hamoudi	

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long – axis anticline with N – S trending with approximate dimensions about 20^{km} width & 60^{km} length near to Iran – Iraq border, to the east of Yaran Oil Field.

The trap area of the South Azadegan is limited to 43.2 x 17.2^{km}. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of $4{,}643^{m}$. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of $5{,}340^{m}$ (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 & AZN-10). Clastic oil – bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil – bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest & AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N – S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus & Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist – bearing packstone & partially grainstone, which contains abundant bioclasts. Some other facies like matrix – dominated wackestone & mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province & is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran – Iraq Border. It is in The Northwest of The YADAVARAN Field & in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer & Highest Temperature is 52$^{°c}$. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

This Document covers the work over of well AZNS – 116. Main headlines of this program are de – complete the well and run new completion string.

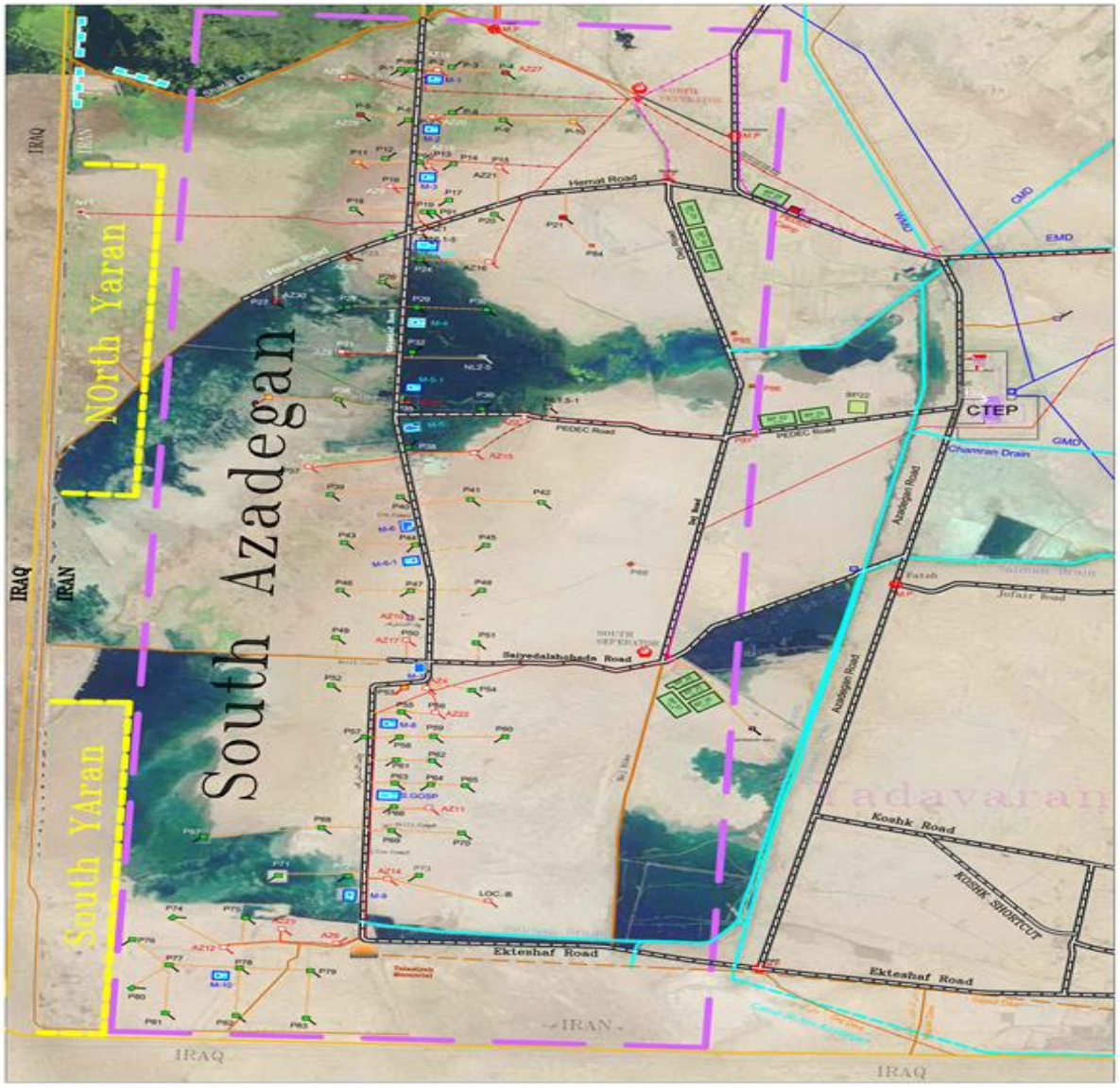


Figure 1: South Azadegan Oilfield

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Workover
Well Name / Type :	AZNS – 116 / Fahliyan well in Block III
Pad Number:	81
Well Type:	Vertical
Wellhead Coordinate:	N: 3,434,999.0 E: 187,374.0
TD Coordinate:	N: 3,434,999.0 E: 187,374.0
Well Objective:	Based on unsuccessful fishing operation results, the decision has been made to De – complete the well, perforate new intervals, change completion string & Re – complete the well.
RTE:	14.45^{m}
GLE:	3.95^{m}
Well Duration :	89 (Main Plan) + 45.2 (Contingency Plan)
Total Depth (MD):	4150^{m}
Producer Formation:	Fahliyan
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1: AZNS-116 Well Information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.

Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: AZNS-116 Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN – 010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN – 010 & AZN – 011. In this block, Gachsaran formation is characterized by pressure tending to normal.

Block III is located on the south of line well AZN – 011. In this block, Gachsaran formation is characterized by normal pressure & all drilled wells in this block suffered from complete loss at Pabdeh formation.

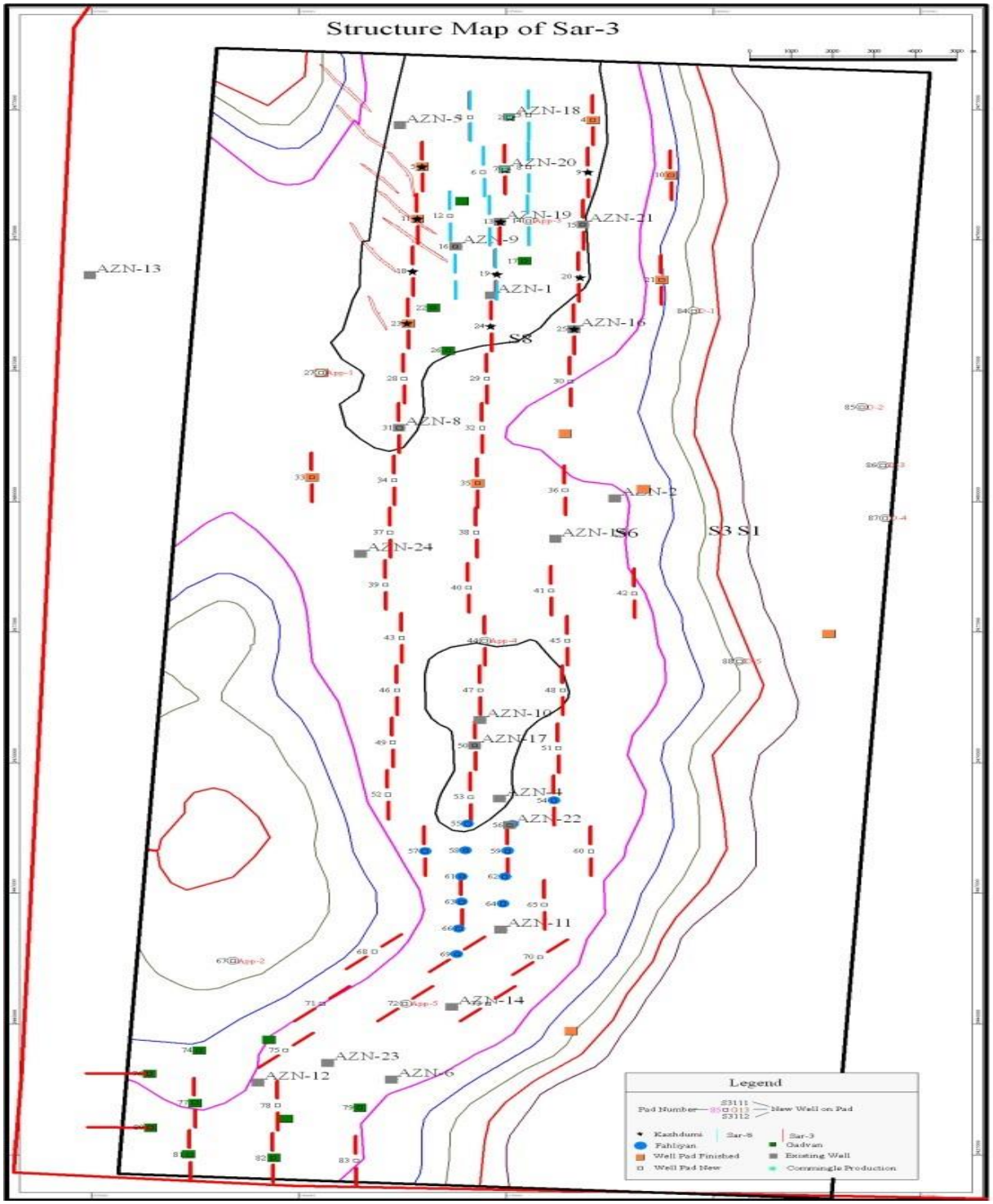


Figure 2: South Azadegan Area Distribution

1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 * 100	25 * 20 * 2	7.5L * 3.80W * 3.40D	30"
Double Well	145 * 100	25 * 20 * 2	7.5L * 3.80W * 3.40D + 4.0L * 3.8W * 3.4D	30"
Triple Well	160 * 100	25 * 20 * 2	7.5L * 3.80W * 3.40D + 4.0L * 3.8W * 3.4D + 4.0L * 3.8W * 3.4D	30'

Table 3: AZNS-116 Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 $^{kg/cm2}$

- Fuel tank concrete pond dimension: Length 20^{m} * Width 12^{m} * H eight 0.5^{m}.
- Drainage: Width 30cm * Depth 30cm.
- Burning pit: 15m * 15m square, located on South – East or South – West with at least 350m distance from well head.
- Cellar to cellar distance in multi – well well pad: 15m.
- Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
- Mud material dimension is 25m * 35m + granular surface.
- Mud drying dimension is 25m * 35m + concrete surface.
- Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL.

1. Types of well pads:

- One well in one pad.
- Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well.
- Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.

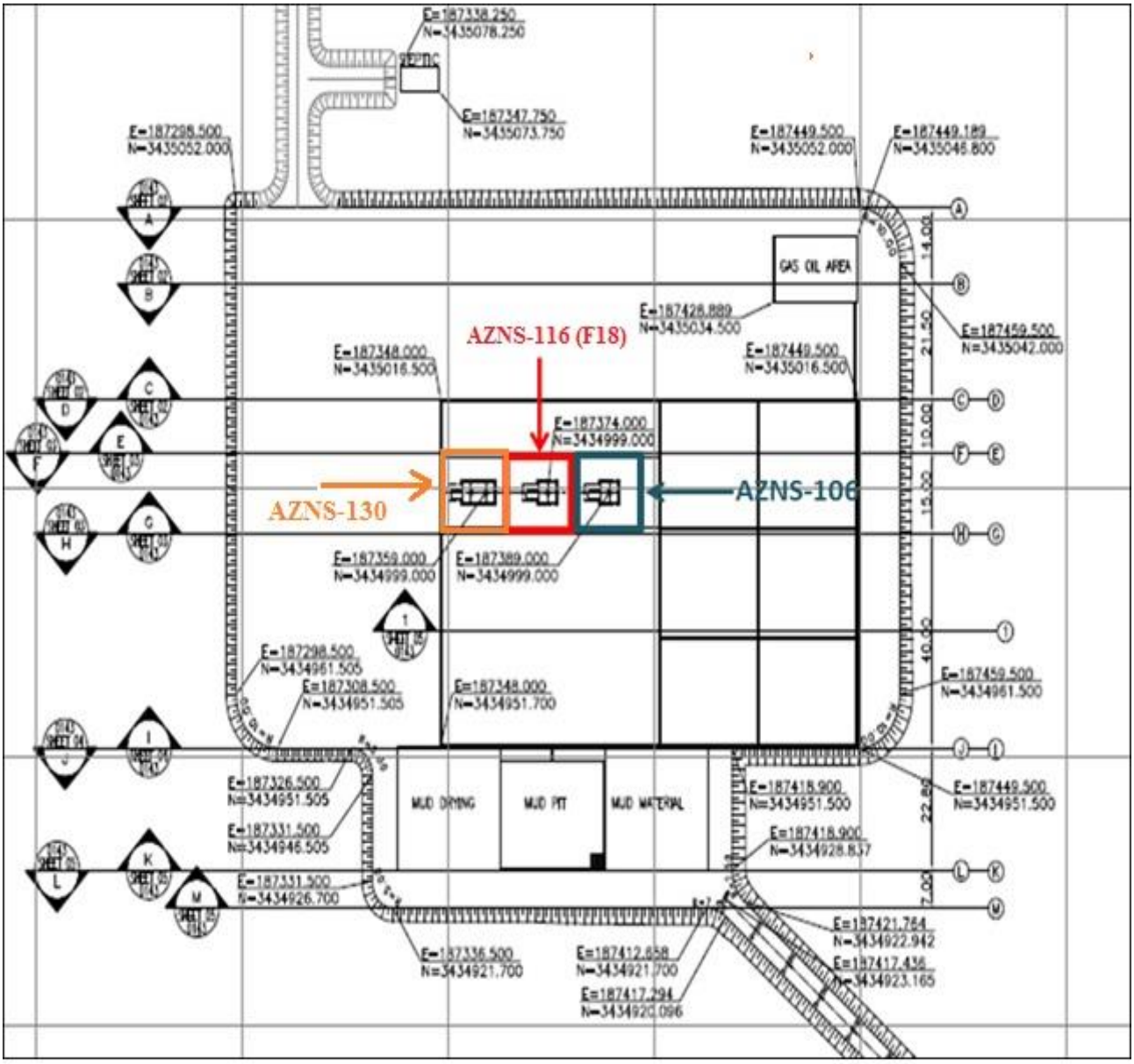


Figure 3: Well Pad Sketch

1.5 Drilling History

26" Hole Section

Rig 304 was ready to spud on 1395/04/11 but due to unavailability of 26” hole equipment rig spud was postponed to 1395/04/14.

Rig 304 was spudded on 1395/04/14 at 12:00 &drilling 26” hole section commenced to TD at 150^{m} while bit balled up was experienced several times & ran totco at 70^{m} & 140^{m} which both survey result showed inclination of 0 degree.

Then ran 12 joints 20” casing, 94^{ppf} K – 55, BTC, Nippon steel, Japan (shoe: 94^{ppf}, K-55, BTC, stab in type, Kherad Sanat) to 146^{m} & cemented by stinger without any major problem. installed 20 – 3/4" 3k CHH (MSP DRILEX) with 2 number 2 – 1/16" 5k gate valve on left & right side & set casing slips as well & injected plastic packing & tested the same with 400^{psi}, Ok.

After nipple up BOP (21 – 1/4" – 2k drilling spool & 21 – 1/4" double rams (top & blind rams, 21 – 1/4" 2k annular preventer) & BOP test, ran in hole with 17 – 1/2” rock bit & drilled out hard cement & shoe to 150^{m} while working on junk sub several times.

17 – 1/2" Hole Section

17 – 1/2” hole section was started on 1395/04/18 & drilling commenced from 150^{m} to 170^{m} while working on junk sub several times. (Recovered 4^{lbs} junk)

Drilling of 17 – 1/2” hole section was continued from 170 to 1327^{m} while at 340^{m} displaced hole with 70^{pcf} PHPA mud & also due to increase losses from 4 to 15^{bph} at 1275^{m} & 1298^{m} pumped 30^{bbl} 20^{ppb} hi-vis LCM pill (10^{ppb} mica+10^{ppb} walnut), losses decreased to 4 – 5BPH & continued drilling 17 – 1/2” hole section from 1327^{m} to 1535^{m}, due to increase losses from 5 to 90BPH, stopped drilling & pumped 30^{ppb} hi-vis LCM pill (15^{ppb} mica+15^{ppb} walnut) in several stage (totally pumped 100^{bbl} LCM pill), losses decreased from 90 to 60BPH. Continued slow circulation & pumped 30^{bbl} 30^{ppb} hi-vis LCM pill (15^{ppb} mica+15^{ppb} walnut) in two stages (totally pumped 160^{bbl} LCM pill) with 60BPH loss. Then Spotted 50^{bbl} 40^{ppb} hi-vis LCM pill (20^{ppb} mica+20^{ppb} walnut) & pulled out from 1535 to 1480^{m} while slowing circulation through string. Losses decreased from 60 to 5^{bph}.

Continued drilling 17 – 1/2" hole size section from 1535 to 1990^{m} with 10 – 2BPH losses while pumped 30^{ppb} hi-vis LCM pill (15^{ppb} mica+15^{ppb} walnut) periodically. (At 1966^{m} Due to increasing of Ca++ from 800 to 1300$^{mg/l}$ increased Mw from 77.5 to 81^{pcf}).

Continued drilling 17 – 1/2" hole section from 1990[m] to 13 – 3/8" casing shoe at 2034[m] w/ 3 – 4BPH loss. After condition trip RIH with wire line CDR/GR logging tool string to 2020[m] (14[m] off bottom) & logged up from 2020[m] to 146[m] & continued POOH to surface & laid down tools.

After 12 days waiting on receiving 13 – 3/8" casing performed condition trip & ran 13 – 3/8" as following:

- Casing 13 – 3/8" float shoe 68[ppf], N-80, BTC
- 2 joints 13 – 3/8" casing 68[ppf], N-80, BTC
- 13 – 3/8" float collar 68[ppf], N-80, BTC
- 178 joints 13 – 3/8" casing 68[ppf], N-80, BTC TO 35

As cement job, pumped 40[bbl] chemical wash, 80[bbl] 85[pcf] spacer & mixed & pumped 246[sxs] cement class G" Delijan (SW: 85[pcf], SV: 150[bbl]) as pre-flush, mixed & pumped 3762[sxs] cement class "G" Delijan (SW: 98[pcf], SV: 1471[bbl]) as lead cement, mixed & pumped 741[sxs] cement class "G" delijan (SW:118[pcf], SV:181[bbl]) as tail cement. Dropped top plug (Kherad Sanat Ariya), displaced with 987[bbl] mud & bumped plug with 1500[psi] & kept for 15min as CIT. during cement job got 700[bbl] slurry cement on surface & recorded totally 250[bbl] loss.

RIH with 12 – 1/4" R. bit & 8 – 1/2" Junk sub & tagged top of cement at 1990[m]. Commenced drill out from 1990 to 2028[m] while D/O float collar at 2009[m] & worked on junk sub several times. Continued D/O hard cement & 13 – 3/8" casing shoe at 2034[m].

12 – 1/4" Hole Section

Drilling of 12 – 1/4" hole section started on 8/23/2016 (1395/06/02) & drilled from 2034 to 2044[m], to change BHA & cement bond logging POOH to surface & L/D 8 – 1/2" junk sub, recovered: 3.5[lb] junk.

RIH with CBL/VDL/CCL/GR logging tool string to 2027[m] (6[m] above shoe) & logged up to 136[m], POOH to surface.

Resumed drilling 12 – 1/4" hole section from 2044 to 2251[m] with 2 – 3BPH loss while continuously pumped 30[bbl] hi-vis pill to sweep hole. At 2251[m] mud loss increased to 70BPH, reduced pumping flow rate & spotted totally 100[bbl] 10[ppb] hi-vis LCM pill, loss decreased to 15BPH.

Continued drilling 12 – 1/4" hole size section to 2303[m] & again due to increase loss from 10 to 50BPH in interval of 2255[m] to 2274[m], pumped 15[ppb] hi-vis LCM pill several times. Loss decreased from 50 to 15BPH. Cut mud weight from 80 to 79[pcf] & loss decreased from 15 to 5BPH.

Continued drilling 12 – 1/4" hole section from 2303 to 2364^{m}, while pumping 30^{bbl} hi vis pill periodically w/3-5BPH loss & experienced high & abnormal torque. Ran Gyro to 2335^{m} & took surveyed (depth: 2335^{m}, dev.:1.11degree, azimuth: 170.8 degree) & at 2135^{m} (dev.:0.42 degree, azimuth: 330 degree).

POOH to surface & RIH with new bit & junk sub & resumed drilling 12 – 1/4" hole with 2364^{m} to 2375^{m} w/2 – 3BPH loss while worked on junk sub several times. POOH to surface.

RIH with same used bit & resumed drilling 12 – 1/4" hole size section from 2375^{m} to 2385^{m}, due to low ROP POOH to surface & due to unavailability of 12 – 1/4" PDC bit, RIH with 12" used PDC bit & 8" PDM to bottom at 2385^{m} & commenced performance drilling as 12" hole size from 2385 to 2648^{m} w/1 – 5BPH loss. Due to low progress POOH to surface & RIH with used 12 – 1/4" b. bit.

Enlarged 12" hole to 12 – 1/4" size from 2385^{m} to 2648^{m} & resumed drilling 12 – 1/4" hole to 2817^{m}. Due to torque fluctuation & low ROP POOH to surface while ran Gyro at 2780^{m} (well deviation: 0.23 deg, azimuth: 106.4 deg).

RIH with new 12 – 1/4" PDC bit & resumed drilling from 2817^{m} to 3200^{m} while pumping hi-vis pill periodically w/3 – 4BPH loss. Performed condition trip, RIH to bottom at 3200^{m} & ran Gyro 7 took survey each 100^{m}. (3100^{m}, incl.: 0.99 deg, azimuth: 178.10 deg-3000^{m}, incl. : 1.04 deg, azimuth: 153.40 deg-2900^{m}, incl.: 1.15deg, 166.70 deg-2800^{m}, incl.: 0.95deg, azimuth:181 deg- 2700^{m}, incl. :1.82 deg, azimuth : 201 deg-2600^{m}, incl.:1.40 deg, azimuth:212 deg-2500^{m}, incl. : 1.44 deg, azimuth:179 deg-2400^{m}, incl. : 0.44 deg, azimuth:222.5 deg).

Continued drilling 12 – 1/4" hole from 3200 to 3490.5^{m} w/3 – 6BPH loss. At 3490^{m} pressure decreased from 2750 to 2400^{psi}, checked surface equipment-ok & decided to POOH to check string. At surface observed 8 – 1/2" DC twisted off from 10^{cm} under box (left in hole:12 – 1/4" b. bit, 8 – 1/2" b. sub, 1 joint 8 – 1/2" DC).

RIH with 11 – 3/4" overshot (assembled with 8 – 3/4" mill control & 8 – 1/2" basket grapple), string got stuck at 2043^{m}, worked on stuck several with max over pull 120^{klb} & jarring up, string got free, POOH to surface 7 laid down overshot, RIH with 12 – 1/4" bit to condition hole. POOH to Surface, RIH with 10 – 1/8" over shot (assembled with 8 – 5/8" control & 8 – 1/2" spiral), engaged fish &d POOH to 3477^{m} with tight hole, string got stuck, worked on stuck pipe with applying 100^{klb} over pull & jarring up several times while pumping salt saturated water (caustic pill) & hi-vis pill, string got free, POOH to surface, recovered the fish completely. RIH with 12 – 1/4" PDC bit to 2055^{m} & continued RIH with hard W&R to 2907^{m}, due to full gauge size of 12 – 1/4" string stabilizer & inability to continue RIH, POOH to change string stabilizer to surface & L/D 12 – 1/4" string stabilizer.

RIH with 12 – 1/4" PDC bit to bottom at 3490.5^{m} & resumed drilling 12 – 1/4" hole from 3490.5 to 3669^{m} w/3 – 6BPH loss. Continued drilling from 3669 to 3671^{m} while due to increase loss from 3 to 30BPH, pumped 15^{ppb} hi-vis LCM pill (mica & wall nut) in 2 stage, loss decreased to 6 – 8BPH.

Continued drilling 12 – 1/4" hole from 3671 to 3730^{m} while increased mud weight from 82 to 84^{pcf} & continued drilling from 3730 to 3885^{m} (9 – 5/8" casing shoe) w/6 – 4BPH loss. POOH to surface.

Ran full suite logging tool string (first run, MSFL/DAL/DLL/CDR/GR) to bottom at 3885^{m}, logged up from 3885 to 2033^{m} (13 – 3/8" casing shoe), & continued POOH with logging tools to surface. To condition hole prior to second logging, RIH with 12 – 1/4" PDC bit to bottom while increasing mud weight from 84 to 85^{pcf}. POOH to surface.

Run full suite logging tool string (second run, CNT/ZDL/NGS) to 3300^{m}, due to missed signal from TCS tools, POOH to surface & change TCS tools, RIH bottom at 3885^{m}, logged up from 3885 to 3400^{m} while circulating cross well head w/1 – 2BPH loss. Again recorded resistivity log data as client request from 3885 to 2033^{m}.

Rigged up full suite logging tool string (third run, MSFL/DLL/GR) to 2000^{m}. Due to GR failure, POOH to surface, changed GR tools & RIH with same logging tool string to bottom at 3885^{m}, logged up from 3885 to 2033^{m} (13 – 3/8" casing shoe), continued POOH with logging tools to surface, L/D logging tool string.

After 26 days wait on receiving 9 – 5/8" casing, ran 9 – 5/8" Casing as following:

- 9 – 5/8" F. shoe (Nasr Mobtakeran, 53.5#, T95, TSH)
- 3 joints 9 – 5/8" casing 53.5#, T-95, TSH blue as shoe track
- 9 – 5/8" F. Collar (Nasr Mobtakeran, 53.5#, T-95, TSH)
- 341 joints 9 – 5/8" Casing 53.5#, T-95, TSH blue

Pumped 36^{bbl} pre-flush & 10^{bbl} 85^{pcf} spacer, mixed & pumped 2465^{sxs} cement "G" delijan with cenosphere (slurry volume:1033^{bbl}, slurry weight:92^{pcf}) as lead & mixed & pumped 312^{sxs} cement "G" delijan with o-silica flour (slurry volume:100^{bbl}, slurry weight:110 – 118^{pcf}) as tail (due to pump truck failure, pumped 16^{bbl} less than program), dropped top plug (Nasr Mobtakeran Co.), bumped plug w/3000^{psi} for 15min, Ok. While after displaced 702^{bbl} got cement slurry on surface (dumped 151^{bbl} slurry cement on surface) & totally recorded 21^{bbl} loss & 40^{bbl} contaminated cement, checked back flow, Ok. RIH with 8 – 1/2" bit to 3848^{m} & D/O plugs & float collar at 3848^{m} & hard cement to 3881^{m}, displaced hole with 100^{pcf} mud 7 continued D/O hard cement, shoe track 7 shoe to 3885^{m} (float shoe at 3882^{m}) while worked on junk sub several times.

8 – 1/2" Hole Section

Commenced drilling 8 – 1/2” hole section from 3885 to 3888^{m}, performed FIT by applying 620^{psi} (EMW: 107^{pcf}) pressure step by step & hold for 15min, Ok. POOH to surface & recovered 1^{lb} junk. RIH with 9 – 5/8” casing scrapper & scraped from 3760 to 3881^{m}, POOH to surface & rigged up logging equipment.

Ran SBT/VDL/CCL/GR logging tool string to 3885^{m}, logged up from 3885 to 2025^{m} (8^{m} above casing shoe 13 – 3/8") & continued POOH to surface. RIH with 8 – 1/2” bit &continued drilling to TD at 4150^{m} while experienced no major drilling problems on this hole section.

Condition hole prior logging, POOH to surface, RIH with run 1st full suite log as: MSFL/DDL/SONIC/CALIPER/CDR/GR tool string to bottom at 4150^{m} with good hole condition, logged up from 4150^{m} to 9 – 5/8” casing shoe at 3882^{m} & continued POOH from 3882^{m} to surface.

Ran 2nd full suite log as: ZDL/CNT/SLT tool string to 3890^{m} (8^{m} below show), string held up at 3890^{m}, tried to pass this point several times, no success, POOH to 3887^{m}, tool string stuck at 3887^{m}, worked on stuck with 1500^{lb} overpull, string got free, continued POOH from 3887^{m} to surface.

RIH with 8 – 1/2” bit & condition hole prior run 7” liner, ran 347^{m} 7" liner as:

- 7" float shoe 29^{ppf}, 110S, TPCQ (Pars Poolad Aghaei Co.)
- 2 joints 7" liner 29^{ppf}, 110S, TPCQ
- 7" float collar 29^{ppf}, 110s, TPCQ (Pars Poolad Aghaei Co.)
- 1 joint 7" liner 29^{ppf}, 110s, TPCQ
- 7" LDC 29^{ppf}, 110s, TPCQ (Pars Poolad Aghaei Co.)
- 27 joint 7" liner 29^{ppf}, 110s, TPCQ
- 7" Hydraulic liner hanger 29^{ppf}, 110S, TPCQ (Pars Poolad Aghaei Co.)
- POB, 29^{ppf}, 110S (Pars Poolad Aghaei Co.)
- Tie back sleeve (Pars Poolad Aghaei Co., ID: 7 – 3/8")

Circulated btm's up, observed hydrocarbon on shakers & got cut Mw from 105 to 98^{pcf} & 150^{bbl} contaminated mud. Continued circulation & balanced mud at 105^{pcf}. Set hydraulic liner hanger, release setting tool. L/U cement line & pumped 9^{bbl} pre-flush & 7^{bbl} 85^{pcf} spacer, mixed & pumped 169^{sxs} cement class "G" Delijan with silica flour (slurry weight: 118^{pcf}, slurry volume: 53^{bbl}), dropped pump down plug,

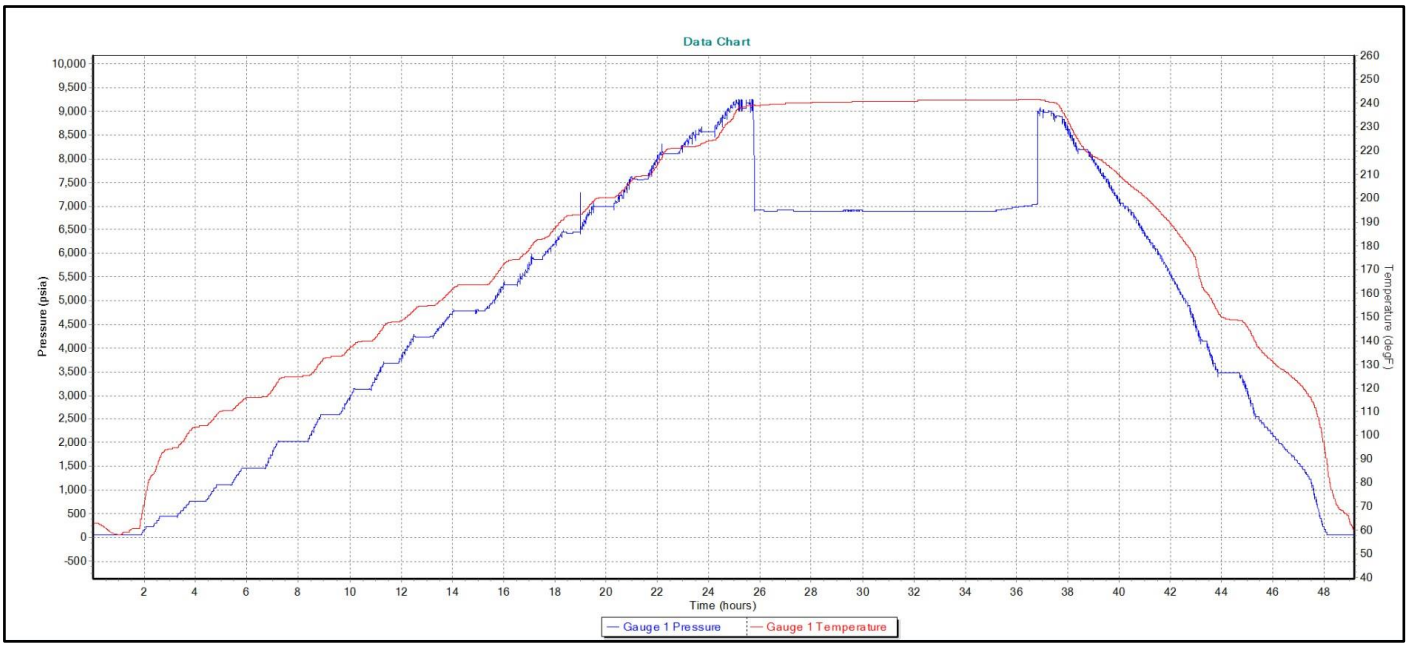
displaced w/259^{bbl} 105^{pcf} mud, Obs. shearing after 221^{bbl} w/1600^{psi} & bumped plug w/3000^{psi} for 15min, Ok, released pressure, checked back flow, Ok.

POOH to surface, RIH with 8 – 1/2" bit to 3695^{m}, tagged cement with 20^{klb} & 350gpm, Ok, D/O medium to hard cement from 3695 to 3802^{m} while confirming 7" liner lap at 3802^{m}. POOH to surface, P/U & M/U 6 – 1/8" bit 7 RIH to 4090^{m}, tagged cement with 20^{klb} & 350gpm, Ok. POOH to surface while performing pressure test against 7" liner lap at 3700^{m} w/2000^{psi} for 15min, Ok. RIH with 9 – 5/8" scraper & scraped from 3650 to 3800^{m}. POOH to surface.

Dry Test

RIH with dry test tools while filling up string with water cushion as per program (KCl mud, 85^{pcf}) to 3770^{m}. Tried to set packer at 3770^{m} several times, no success. POOH to 3760^{m}, revers circulated & got 205^{bbl} 85^{pcf} mud, POOH to surface & L/D 9 – 5/8" dry test tools, worked on dry test packer. RIH with same dry test tools to packer depth at 3780^{m} while filling up 3557^{m} of string w/85^{pcf} KCl mud as per program to 3785^{m}.

Set packer at 3780^{m}, opened hyd. spring valve for 3hrs, no bubble observed on surface & monitored annulus, static, closed DCIP for 6hrs meanwhile observed the annuls, static. Dropped bar & shear reveres sub, revers circulated, got 207^{bbl} 197^{bbl} 85^{pcf} KCl mud & 10^{bbl} 75^{pcf} hi-vis pill, circulated & balanced mud weight at 105^{pcf}, unset packer & observed well, static. POOH to surface & L/D test tools.



Oil Exploration Operations Company MDST RTTS Report		TDDC
Job: DRY TEST	Region:	Operation Date: 25/12/12
Filed: AZAS	Well: 116	Rig: TDDC304

Set Packer Depth:			Water Cushion:	
Middle B.T. Depth:			Bottom B.T. Depth:	
Row	Discription	Buttom Bt.	Middle Bt.	Comment
1	Initial Mud Static Pressure	9108	9105	
2	Initial Flow Pressure	4909	4901	
3	Final Flow Pressure	4199	4194	
4	Close Pressure	7038	7035	
5	Final Hydrostatic Pressure	9057	9049	

Plug & Suspension

Due to unavailability of completion string decided to plug back 7 temporarily suspend well AZNS-116. To set 1st cement plug RIH w/open ended DPs to top of cement at 4090m. Added 0.5% volume corrosion inhibitor & 0.5% volume bio side to mud system & adjust PH on 10.5. Then pumped 7bbl salt saturated mud ahead as spacer, mixed & pump 75sxs cement class "G" delijan (slurry weight.:118pcf, slurry volume: 24bbl), pumped 0.9bbl salt saturated mud as behind spacer then displaced w/192bbl mud as balance plug, After 15 hrs wait on cement RIH & tagged cement at 3434m, washed down to 3440m, & tagged hard cement with 450gpm & 20klb, POOH to 3400m, performed pressure test against cement plug w/2000psi for 20min, Ok, POOH from 3400 to 2030m.

At 2030m to set 2nd cement plug, L/U cement line & pumped 7bbl salt saturated mud ahead as spacer, mixed & pump 99sxs cement class "G" delijan (SW: 118pcf, SV: 24bbl), pumped 1bbl salt saturated mud as behind spacer then displaced w/99bbl mud as balance plug, after 15hrs wait on cement, RIH & tagged soft cement at 1948m, washed down & tagged hard cement at 1949m with 450gpm & 20klb. POOH to surface while performing pressure test against 2nd cement plug w/2000psi for 20min, Ok at 1890m.

N/D 13 – 5/8" BOP stack, installed 11" 10,000psi blind flange (Pars Poolad Aghaei) & tested w/2000psi, Ok. Released rig on: 95/09/27 at 20:00.

Work-Over#01

Spudded rig 303 on 1396/08/02 at 00:00, ran in hole to 1951^{m}, tagged upper cement plug, circulated mud & adjusted mud properties & mud weight to 105^{pcf}, drilled out medium to hard cement plug from 1951^{m} to 2018^{m}, washed down from 2018^{m} to 2040^{m}, pumped 30^{bbl} hi-vis pill, circulated mud & adjusted mud properties & mud weight to 105^{pcf}, observed the well – static, continued run in hole to top of second cement plug at 3440^{m}, circulated mud & adjusted mud weight to 105^{pcf}, drilled out medium cement from 3440 to 3472^{m}, washed down from 3472 to 3560^{m}, continued drill out medium to hard cement from 3560 to 3790^{m} while pumped hi-vis pill periodically with NL-NG, continued run in hole to 7" liner lap at 3804^{m}, pumped 30^{bbl} hi-vis pill & circulated & uniform mud properties & adjusted mud weight to 105^{pcf} with NL-NG, suddenly pressure decreased from 1860 to 1420^{psi}.

Pulled out to surface, ran in hole with 7" scraper to top of cement at 4090^{m} while scraped 7" liner, circulated mud & increased mud weight from 105 to 106^{pcf}, spot 35^{bbl} 84^{pcf} $CaCl_2$ at bottom to 7" liner lap, observed the well – static, pulled out to surface & laid down DPs.

To run cement evaluation log of 9 – 5/8" Casing, rigged up Sealand logging equipment & ran in hole with CCL/GR/SBT/VDL logging tool string to 1^{m} above top of Cement at 4089^{m}, logged up from 4089^{m} to 3770^{m}, pulled out to surface & laid down logging tools, ran in hole with NGS logging tool string to 1^{m} above top of cement at 4089^{m}, logged up from 4089 to 3750^{m}, pulled out to surface & rigged down logging equipment.

DST – TCP: 1st TCP Run

After 70hrs wait on receive Explosive, loaded 7 No. 4 – 1/2" HMX TCP guns, totally 10^{m}, 72 degree phasing & 5SPF, arranged to perforate from 4001 to 3991^{m}, ran in hole with DST – TCP string to 4003.75^{m} (bull nose depth) while filled up string with 80^{pcf} $CaCl_2$ as cushion & passed drift & pressure tested at 90^{m}, 1000^{m}, 2000^{m} & 3000^{m} with 6000^{psi}, Ok, rigged up logging equipment & to correlate depth, ran GR/CCL tool string to 3923^{m}, confirmed RA marker depth. Pulled out to surface & laid down equipment, space out, installed flow head & connected to chock manifold & pressure tested same with 6000^{psi}, Ok, by 6 right turn & applying 20^{klb} weight set packer 3952.87^{m}.To perform perforation job, closed the well, applied 1800^{psi} pressure into annulus to open TTV, bleed off pressure to 1000^{psi}, fired guns by applying 7000^{psi} pressure in string several times, no success, to check string & function test of TTV & POTV, unset packer, closed the well & applied 2000^{psi} pressure in annulus, no pressure indication inside string on chock manifold, closed the well & applied 2000^{psi} pressure in annulus, no pressure indication inside string on chock manifold, bleed off pressure & again applied 2200^{psi} pressure in annulus, suddenly string pressure increased to 3300^{psi} & detected indication that POTV became opened, bleed off annulus pressure to zero & opened the well, set packer, applied 1000^{psi} in annulus while observed inside string pressure constant at 3300^{psi}, increased annulus pressure to 2000^{psi}, to fire guns applied 8000^{psi} pressure in string, no success, released annulus pressure to zero, opened the well, unset packer, by opening chock manifold valve tried to release string pressure, no success, set packer, again tried to release

string pressure, no success & detected indication that AORV (secondary reversing valve) in open position, closed the well & to balance mud weight reverse circulated with Max 500^{psi} pressure from annulus meanwhile got 200^{bbl} 80^{pcf} $CaCl_2$ & dumped 40^{bbl} contaminated mud, opened the well, unset packer with Max 35^{klb} over pull, laid down flow head & observed the well – static.

Pulled out with TCP – DST to 50^{m} meanwhile checked tools, observed primary circulation valve in closed position, tubing fill test valve in opened position, pressure operated tester valve in opened position, function tested the same, Ok, secondary tester valve in opened positon, checked fire head & observed all pins sheared & first cord on blank gun was operated led to bullets misfired. Due to misfired guns, waited on daylight while slow circulated with NL- NG. Pulled out with TCP – DST string from 50^{m} to surface, all fire head pins sheared & first cord on blank gun was operated & blank gun detonator cord on gun #5 was disconnected from booster, retrieved all bullets & detonator cord.

DST – TCP: 2nd TCP Run

Loaded 7 No. 4 – 1/2" HMX TCP guns, totally 10^{m}, 72 degree phasing & 5SPF arranged to perforate from 4001 to 3991^{m}, ran in hole with TCP – DST string to 3942^{m} while filled up string with 80^{pcf} $CaCl_2$ as cushion & passed drift & pressure tested at 90^{m}, 1000^{m}, 2000^{m} & 3000^{m} with 6000^{psi}, Ok, at 3942^{m} observed shock & vibration on string (no drag was experienced) & string weight increased from 210^{klb} to 330^{klb} on weight indicator, continued run in hole with 20^{klb} drag down to 3950^{m}, tried to continue run in hole, observed 50^{klb} drag, attempted to pull out with Max 30^{klb} over pull, suddenly observed u – tub from annulus to inside string, monitored string pressure – constant at 2300^{psi}, closed the well & reverse circulated with Max 1000^{psi} pressure from annulus & applied 300^{psi} as back pressure from choke manifold, could not balance mud weight due to shortage of mud on surface & got 147^{bbl} $CaCl_2$ as return & dumped 160^{bbl} contaminated mud, opened the well & observed well – static, to check string & equipment pulled out with TCP – DST from 3950^{m} to surface, checked equipment, MRCV in close position, TTV in open position & AORV in close position, FIS in open position, all 14 pins of fire head sheared & guns fired completely.

Rigged up logging equipment & to check perforation interval ran in hole with multi finger Caliper/GR/CCL tool string to 3932^{m}, tool string held up, tried to pass, no success, logged up from 3932 to 3700^{m}, pulled out wire line to surface.

To check 3932^{m} (the point of logging tool string held up) ran in hole with 6" Metal muncher junk mill to 3932^{m}, circulated mud, ran in hole with soft ream from 3932 to 3970^{m} while checked interval stand by stand at several time – found no obstruction, continued run in hole to 4090^{m}, circulated mud, spotted $CaCl_2$ from 4090^{m} to 7" liner lap at 3802^{m} & observed the well – static.

Pulled out to surface, rigged up logging equipment to check & confirm perforated interval ran in hole with multi finger Caliper/GR/CCL to 3700^{m}, tool string held up & tried to pass several times, no success, pulled out to surface & observed finger probe failure, again ran in hole with same logging tool string to 3700^{m}, tool string held up & tried to pass several times, no success, pulled out to surface & again found

finger probe failed, for 4th time ran in hole with GR/CCL logging tool string to 4050m, logged up from 4050 to 3850m, continued pull out to surface & laid down logging tools.

NOTE:

4th log result showed wrong interval was perforated by TCP, the confirmed perforated depth: 3930m – 3940m.

To double check perforated interval ran in hole with GR/CCL to 3700m, tool string held up & tried to pass several times, no success, pulled out to surface. Ran in hole with 6 – 1/8" b. bit & 7" scrape to 3930m meanwhile scrape inside 9 – 5/8" casing & 7" liner, pulled out to surface. To re – check perforation interval ran MTT/GR to 4050m, logged up from 4050 to 3880m, pulled out with wire line to surface & laid down logging tools. Ran in hole with 6" Metal muncher junk mill to 7" liner lap at 3802m & circulated mud with NL-NG while wait on decision. Decided to perforate planned interval on WCP, ran in hole from 3770 to 4090m & spotted 45bbl 80pcf $CaCl_2$, observed the well – Static, pulled out to surface.

WCP: 1st WCP Run

Rigged up Asmari wireline equipment, ran in hole with 4 – 1/2" HMX gun, 6m, 83 charge, 72 degree phasing, 5SPF arranged to perforate from 4001 to 3996m with CCL to 4030m while correlated depth at 7" liner lap & correlated perforation interval, fired guns, checked perforation interval with CCL, confirmed perforated interval as plan, 4001 – 3996m, pulled out to surface, all 83 No. bullets were fired.

WCP: 2nd WCP Run

Rigged up Asmari wireline equipment, ran in hole with 4 – 1/2" HMX gun, 6m, 83 No. bullet, 72 degree phasing, 5SPF arranged to perforate from 3996 to 3991m with CCL to 3996m while correlated depth at 7" liner lap & correlated perforation interval, fired guns, pulled out to surface, all 83 No. bullets were fired.

To check perforated depth ran in hole with MTT logging tool string to 4030m, logged up from 4030 to 3880m, ran back to 4030m, logged up from 4030 to 3880m, pulled out to surface.

NOTE:

As per MTT log result & interpretation, wrong interval 3913.5 to 3918.5m was perforated, the planned interval was from 3996 to 3991m.

WCP: 3rd WCP Run

To perforate planned interval ran in hole with 4 – 1/2" HMX gun, 6m, 83 No. bullet, 72 degree phasing, 5SPF arranged to perforate from 3996 to 3991m & CCL to 7" liner lap 3810m, CCL showed noise, tool string held up, after several attempts passed the point, ran in hole to 3818m, tool string held up, tried to

pass several times, No success, pulled out to surface, laid down perforation tools, rigged down Asmari wireline equipment, ran in hole with 6" junk mill & 7" Scraper to 4050^{m} while scraped from liner lap at 3802 to 3910^{m} several times, pulled out to 3780^{m} & waited on receive perforation head to rig site meanwhile circulated mud with 0 – 1bph loss, ran in hole to 4050^{m} while scraped 3802 to 3910^{m}, spotted 42^{bbl} 80^{pcf} $CaCl_2$, observed the well – static, pulled out to surface.

WCP: 4th WCP Run

Rigged up Asmari wireline equipment, ran in hole with 4 – 1/2" HMX gun, 6^{m}, 83 bullets, 72degree phasing, 5SPF with CCL to 3996^{m}, correlated depth at 7" liner lap & at planned perforation interval at 3991 – 3996^{m}, fired guns, logged up to 3779^{m} & confirmed perforated interval with CCL successfully as per plan from 3996 to 3991^{m}, pull out to surface, check tools & all bullets fired completely.

To check perforated interval, ran in hole with MTT/GR to 3890^{m}, due to missed signal check tools pulled out to surface, changed Telemetry & ran in hole to top of perforated interval at 3990^{m}, tool string held up, tried to pass – No success, tried to pass the point – No success, logged up from 3990 to 3770^{m}, pulled out to surface.

To clean out inside liner ran in hole with 4 – 1/8" Metal muncher junk mill to 4010^{m} while dressed 3991 to 3995^{m} with Max 2 – 4^{klb} weight, Ok, circulated bottoms up, pulled out to 3780^{m}, circulated mud meanwhile waited on receive DHPG, ran back to 4010^{m}, pumped 80^{pcf} $CaCl_2$ as spacer ahead, displaced inside 7" liner with $CaBr_2$ as completion fluid with 0 – 1bph loss ($CaBr_2$ from 4010 to 3890^{m} & $CaCl_2$ from 3890 to 3700^{m}), pulled out from 4010 to 3890^{m}, displaced hole with $CaBr_2$ as completion fluid till got net completion fluid. As return, totally pumped 927^{bbl}, loss 25^{bbl} & dumped 30^{bbl} contaminated fluid, circulated with completion fluid to balance in & out weight at 106^{pcf} meanwhile added O_2 scavenger & inhibitor to system.

Pulled out from 2200 to 700^{m}, observed the well – 0.1bph gain, closed the well & monitored pressure for 1.5hrs – zero, opened the well & continued pull out to surface. Waited on receive DHPG while rack back 10 stands 3 – 1/2" tubing & 47 stands 4 – 1/2" tubing, ran in hole with 4 – 1/8" Metal muncher junk mill to 3900^{m} while filled up string with 105^{pcf} completion fluid, observed the well – 0.1bph gain, circulated completion fluid & got return with 101 – 105& weight.

Waited on receive DHPG while continued circulation with completion fluid meanwhile added O_2 scavenger & balanced in & out weight at 105^{pcf}, observed the well – 0.1bph gain. To run completion pulled out from 1450^{m} to surface while laid down 262 joint 5" DPs with 0.1bph gain (oil lubricated to surface).

1st Time Completion Running

Rigged up Completion running equipment, ran completion string as per program as below:

- 3 – 1/2" mule shoe 10.2^{ppf}, L – 80, HSM-1, weiqixin
- 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" bottom No-Go landing nipple 10.2^{ppf}, L – 80, HSM-1, Seal bore: 2.562", type: XN, weiqixin , without plug inside
- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1
- XOS 4 – 1/2" HSM pin X TPG2 box
- Mill out extension 18.9^{ppf}, TPG2, Super 13Cr – 110ksi7" hydraulic permanent production packer, model WSDP to be run in 7" 29 – 32^{ppf} casing, 13Cr – 110ksi, Anchor seal 18.9^{ppf}, TPG2, Super 13Cr – 110ksi, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1 to 36^{m} with 0.1bph gain (oil lubricated to surface).

Rigged up OTS slickline equipment, ran 2.8” gauge ring, Ok. Continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" sliding side door, 2.75" check valve plug inside in close position, type X, profile 2.750", 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1
- 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- 3 – 1 /2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" Expansion joint 10.2^{ppf}, L – 80, HSM-1 10^{ft} stroke, in fully open position dressed with 5 No. shear pin, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1
- 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1 to 67^{m}.

Pressure tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8” gauge ring, Ok, continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, Vam top box X HSM-1 pin, 3 – 1/2" DHPG 10.2^{ppf}, Super 13Cr – 110ksi Vam top, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, Vam top box X HSM-1 pin to 72^{m} with 0.1bph gain (oil lubricated to surface).

Connected gauge to cable head, tested all connections, Ok, function tested, pressure tested & checked temperature, all Ok, set pressure gauge in place on gauge mandrel & tested connection, Ok, tested completion string with 8000^{psi} for 15min, Ok, continued run completion string as:

- 1 joint 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" CIV #1 on tubing retrievable chemical injection mandrel 10.2^{ppf}, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" pup joint 10.2^{ppf}, L - 80, HSM-1
- 3 – 1/2" CIV #2 on tubing retrievable chemical injection mandrel 10.2^{ppf}, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run completion string as:

- Ran 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1.

Tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8” gauge ring, Ok, & continued run completion string as:

- 29 joints 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- XOS 3 – 1/2" HSM-1 pin X 4 – 1/2" HSM-1 box

Tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, & ran 2.8” gauge ring, Ok, continued run completion string as:

- 72 joints 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 650^{m} while use cross coupling as per program with 0.1bph gain (oil lubricated to surface).

Tested completion string with 8000psi-ok, continued run completion string as:

- 169 joints 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 2000^{m}

Tested completion string with 8000^{psi}, not Ok, rigged up slickline BOP, lubricator & pressure tested slickline BOP with 3000^{psi}, Ok, ran in hole with pulling tools & retrieve 2.750" check valve, pulled out to surface, laid down BOP & lubricator.

Circulated through string & added O_2 scavenger as per program with 0.1bph gain (oil lubricated to surface), rigged up slickline BOP, lubricator & pressure tested slickline BOP with 3000^{psi}, Ok, ran in hole with 2.750" check valve & set it in SSD, pressure tested of completion string with 8000^{psi}, not Ok, pulled out to surface, changed check valve & packing, ran again & set in SSD, pressure tested completion string with 8000^{psi}, not Ok, pulled out to surface, ran in hole with 2.750" plug & tried to set SSD several times, no success, pulled out to surface, changed plug, ran in hole & set 2.750" plug in SSD,

pulled out to surface, ran in hole with prong & set it in plug of SSD, pulled out to surface, pressure tested completion string with 8000^{psi}, not Ok.

Pulled out from 2000 to 1100^{m} meanwhile pressure tested completion string at 1700^{m}, 1400^{m} & 1100^{m} with 8000^{psi}, not Ok. Rigged up slickline, retrieved plug & prong from SSD, laid down slickline, Pulled out from 1100^{m} to above 2nd CIV at 110^{m}, set check valve in SSD, pressure tested completion string with 8000^{psi}, not Ok, changed check valve packing & set again in SSD, pressure tested string with 8000^{psi}, not Ok, laid down CIV #2, pulled out to above 1st CIV at 80^{m}, pressure tested completion string with 8000^{psi}, not Ok, laid down 1st CIV, continued pull out to above expansion joint at 55^{m}, laid down DHPG, pressure tested above expansion joint with 3000^{psi} & observed leakage from top connection of expansion joint, laid down same.

2nd Time Completion Running

Changed 3 – 1/2" 10.2^{ppf}, L – 80, HSM-1 (10^{ft} stroke in full open position) expansion joint with new one & also changed above & below 3 – 1/2" 10.2^{ppf}, L – 80, HSM-1 pup joint, pressure tested with 8000^{psi}, Ok, rigged up slickline, ran 2.8" gauge ring, Ok & continued run completion string as:

- 1 joint 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup joint, 10.2^{ppf}, L – 80, Vam top box X HSM-1 pin, 3 – 1/2" DHPG, $^{10.2}$ppf, Super 13Cr – 110k, 3 – 1/2" pup joint, 10.2^{ppf}, L – 80, Vam top pin X HSM-1 box

Tested all connections, Ok, function tested, pressure tested & checked temperature, Ok, tested completion string with 8000^{psi}, Ok, continued run completion string as:

- 1 joint 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup joint, 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" CIV #1, 10.2^{ppf}, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup joint 10.2^{ppf}, L-80, HSM-1, 3 – 1/2" CIV #2, tubing receivable chemical injection mandrel, 10.2^{ppf}, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM1

Tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8" gauge ring, Ok, continued run completion string as with 0.1bph gain (oil lubricated to surface):

- 29 joints 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1, XOS 3 – 1/2" HSM-1 pin X 4 – 1/2" HSM-1 box

Tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8" gauge ring, Ok, continued run completion string as with 0.1bph gain (oil lubricated to surface):

- 111 joints 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 930^{m} while used cross coupling

Pressure tested completion string with 8000^{psi}, not Ok, rigged up slickline, ran in hole with pulling tools & retrieved 2.750" check valve, circulated through string to uniform completion fluid while added O_2 scavenger as per program.

Rigged up lubricator & ran in hole with 2.750" check valve, set it in SSD, pressure tested string with 8000^{psi}, Ok, continued run in hole from 2000^{m} as with 0.1bph gain (oil lubricated to surface):

- 109 joints 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam

Pressure tested completion string with 8000^{psi}, not Ok, rigged up slickline, ran in hole with pulling tools & tried to retrieve 2.750" check valve, no success, pulled out to surface, again ran in hole & after several attempts retrieved 2.750" check valve from SSD, pulled out to surface.

Ran in hole with 2.750" plug & set it in SSD, pulled out to surface, ran in hole with prong & set it in SSD plug, pulled out to surface & rigged down slickline, pressure tested completion string with 8000^{psi}, not Ok, rigged up slickline & lubricator, ran in hole & retrieved plug & prong from SSD in 2 runs, ran in hole with 2.750" check valve & set it in SSD, pulled out to surface, rigged down slickline equipment, pressure tested string with 7000^{psi}, not Ok, pulled out from 3000 to 2000^{m} meanwhile pressure tested string at 2500^{m} & 2000^{m} with 8000^{psi}, not Ok, rigged up slickline, ran in hole with pulling tools & retrieved 2.750" check valve from SSD, rigged up slickline equipment, pulled out from 2000^{m} to above CIV #2 110^{m}, rigged up slickline, set 2.750" check valve in SSD, pressure tested completion string with 8000^{psi}, not Ok, continued pull out to above expansion joint at 70^{m}. Laid down both CIV & DHPG, pressure tested completion string with 8000^{psi}, not Ok, observed leakage from expansion joint, laid down expansion joint, pulled out above SSD at 40^{m}, pressure tested with 8000^{psi}, Ok, continued pull out to surface, observed packer rubber element damaged.

Ran in hole with 15 joints 4 – 1/2” tubing to 145^{m} & circulated completion fluid with 0.1bph gain (oil lubricated to surface) while waited on receive new expansion joint.

3rd Time Completion Running

Ran completion string for 3rd time as below:

- 3 – 1/2" mule shoe 10.2ppf, L – 80, HSM-1
- 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM-1
- 3 – 1/2" bottom No-Go landing nipple 10.2ppf, L – 80, HSM-1, seal bore: 2.562", type: XN, without plug inside)
- 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM-1, 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1
- XOS 4 – 1/2" HSM-1 pin X TPG2 box
- Mill out extension 18.9ppf, TPG2, Super 13Cr – 110ksi
- 7" hydraulic permanent production packer, model: WSDP to be run in 7" 29 – 32ppf casing, 13Cr – 110ksi, anchor seal 18.9ppf, HSM-1, Super 13Cr – 110ksi
- 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM1
- 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1, 3 – 1/2" SSD, 2.750" check valve plug inside in & closed, type: X, profile: 2.750", 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1

Pressure tested with 8000psi, Ok, continued run string as:

- 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM-1, 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1, 3 – 1/2" expansion joint 10.2ppf, L – 80, HSM-1, 10ft stroke in fully open position with 4no shear pin 3 – 1/2" pup joint 10.2ppf, L – 80, HSM-1

Pressure tested with 8000psi, Ok, rigged up slickline equipment, ran 2.8” gauge ring, Ok, continued run string as:

- 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM-1
- 3 – 1/2" XOS 10.2ppf, L – 80, Vam top box X HSM-1 pin, 3 – 1/2" DHPG, 10.2ppf, Super 13Cr – 110ksi, Vam top, 3 – 1/2" pup joint 10.2ppf, L – 80, Vam top pin X HSM-1 box
- 1 joint 3 – 1/2" tubing, 10.2ppf, L – 80, HSM-1
- 3 – 1/2" pup joint 10.2ppf, HSM-1, 3 – 1/2" CIV #1, 10.2ppf, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000psi, Ok, continued run completion string as:

- 3 – 1/2" pup joint, 10.2ppf, HSM-1, 1 joint 3 – 1/2" tubing 10.2ppf, L – 80, HSM-1

- 3 – 1/2" pup joint, 10.2^{ppf}, HSM-1, 3 – 1/2" CIV #2 10.2+, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run with 3 – 1/2" pup joint, 10.2^{ppf}, HSM-1

Rigged up slickline equipment, ran gauge ring, Ok, continued run completion string as:

- 3 stands 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1 to 180^{m} while used cross coupling as per program with 0.1bph lubricate, continued run completion string as:
- 7 stands 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- XOS 3 – 1/2" HSM-1 pin X 4 – 1/2" HSM-1 box
- 2 stand 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 450m with 0.1bph gain (oil lubricated to surface)

Pressure tested completion string with 8000^{psi}, not Ok, rigged up slickline 7 retrieved SSD check valve, redressed same, set check valve in SSD, pressure tested, not Ok, to check string pull out from 450^{m} to above CIV #2 at 100^{m}, pressure tested, not Ok, laid down CIV #2, continued pull out to above CIV #1 at 82^{m}, pressure tested with 8000^{psi}, observed leakage from check valve of CIV #1, laid down, waited on receive new CIV meanwhile pull out to above DHPG at 75^{m}, pressure tested completion string with 8000^{psi}, Ok, laid down same & pulled out to 65^{m} above expansion joint, pressure tested completion string with 8000^{psi}, Ok.

4th Time Completion Running

Received new CIV check valve, changed same, to check 7" hydraulic packer pulled out to 25^{m} & checked same, Ok meanwhile laid SSD. Continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" new sliding side door, 2.750" check valve plug inside in close position, type X, profile: 2.750”, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1 to 40^{m}

Pressure tested with 8000^{psi}, Ok, continued run as:

- 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" expansion joint 10.2^{ppf}, L – 80, HSM-1, 10^{ft} stroke – in fully open position with 4no shear pin, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1
- 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM-1

- 3 – 1/2" XOS 10.2^{ppf}, L – 80, Vam top box X HSM-1 pin, 3 – 1/2" DHPG, 10.2^{ppf}, Super 13Cr – 110ksi, Vam top, 3 – 1/2" pup joint 10.2^{ppf}, L – 80, Vam top pin X HSM-1 box
- 1 joint 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1
- 3 – 1/2" pup jt. 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" CIV #1, tubing retrievable chemical injection mandrel, 10.2^{ppf}, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run completion string as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM1
- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 3 – 1/2" CIV #2, tubing retrievable chemical injection mandrel 10.2^{ppf}, L – 80, HSM-1, Super 13Cr – 110ksi with dual check valve & rupture disk, weiqixin

Connected 3/8" control line, tested rupture disk with 2000^{psi}, Ok, continued run as:

- 3 – 1/2" pup joint 10.2^{ppf}, L – 80, HSM-1, 1 joint 3 – 1/2" tubing 10.2^{ppf}, L – 80, HSM1 to 106^{m} with 0.1bph gain (oil lubricated to surface)

Pressure tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8” gauge ring, Ok, continued run completion string as:

- 29 joints 3 – 1/2" tubing, 10.2^{ppf}, L – 80, HSM-1
- XOS 3 – 1/2" HSM-1 pin X 4 – 1/2" HSM-1 box
- 1 stand 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 406^{m}

Rigged up slickline equipment, ran 2.8” gauge ring, Ok, continued run as:

- 37 stands 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 1470^{m} while use cross coupling with 0.1bph gain (oil lubricated to surface).

Continued run completion string as:

- 31 stands 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 2350^{m}

Pressure tested completion string with 8000^{psi}, Ok, rigged up slickline equipment, ran 2.8 gauge ring, Ok, continued run completion string as:

- 156 joints 4 – 1/2" tubing, 18.9^{ppf}, L – 80, New Vam to 3300^{m} while use cross coupling with 0.1bph gain (oil lubricated to surface)

Pressure tested completion string with 8000^{psi}, Ok, continued run completion string as:

- 4 – 1/2" XOS New Vam pin X TPG2 box, 4 – 1/2" flow coupling 18.9^{ppf}, L – 80, TPG2, Super 13Cr – 110ksi, 4^{ft}, weiqixin
- 4 – 1/2" TRSSSV, tubing retrievable safety valve, 18.9^{ppf}, L – 80, TPG2 with 3.437" seal bore, Super 13Cr – 110ksi, weiqixin
- 4 – 1/2" flow coupling, 18.9^{ppf}, L – 80, TPG2 Super 13Cr – 110ksi, 6^{ft}, weiqixin
- 4 – 1/2" pup joint 18.9^{ppf}, L – 80, TPG2, 4 – 1/2" XOS TPG2 pin X 4 – 1/2" New Vam box to 3841^{m}

Connected 1/4" control line to TRSSV, tested control line with 10,000^{psi}, Ok, continued run completion string as:

- 4 joints 4 – 1/2" tubing 18.9^{ppf}, L – 80, New Vam to 3880^{m} while held 4000^{psi} pressure inside control line.
- XOS 4 – 1/2" New Vam pin X 4 – 1/2" TPG2 box
- 4 – 1/2" fluted swage double pin Vam top X TPG2
- Tubing hanger MSP 11" X 4 1/2" ACME box up X 4 1/2" Vam top box down
- XOS 4 – 1/2" ACME Pin X new Vam box to 3883^{m} with 0.1bph gain (oil lubricated to surface)

Tested completion string with 8000^{psi}, Ok & inflow tested while held pressure inside string, released control line pressure to zero, closed TRSSV, bleed off pressure above TRSSV to 500^{psi}, observed pressure for 15min, constant, increased pressure inside string to 6000^{psi} (above flapper) then opened TRSSV flapper with 3700^{psi}, observed pressure inside pressure increased to 7000^{psi}, Ok, released pressure.

Connected 1/4" control line 7 2no 3/8" CIL & 1/4" DHPG line to tubing hanger, picked up 4 – 1/2" landing joint 18.9^{ppf}, L – 80, New Vam & 2no 4 – 1/2" pup joint New Vam, 18.9^{ppf}, L – 80 & ran in hole to 3894^{m}, touch THS, to space out picked up string as 1.5^{m}.

Rigged up slickline, as 1st run; installed 3.437" protection sleeve of TRSSV, as 2nd run; retrieved 2.750" check valve plug from SSD, as 3rd run; installed 2.560" plug & prong in No-Go, as 4th run; opened SSD by shifting tool.

Lined up pump truck to completion string increased press step by step to 4000psi, set 7" hydraulic permanent production packer at 3870m, pressure tested string with 8000psi, Ok, over pull tested packer, Ok, landed tubing hanger in THS & slack off completely string weight while shear expansion joint pins (Expansion joint in closed position as 1.5m), tighten tie down screws, closed rams & tested tubing hanger with 1000psi, Ok, tested completion string with 8000psi, Ok, rigged up slickline equipment, as 5th run: retrieved 2.560" prong from No-Go, retrieved 2.560 plug from No-Go landing nipple, tested annulus with 2000psi, Ok, retrieved protection sleeve from TRSSV landing nipple, function tested TRSSV as open & close, Ok, back off landing joint, installed 2-way BPV, tested two way & tubing hanger body with 7500psi, Ok, nipple down BOP stack, checked hanger neck, Ok, installed MSP bottom master valve, 11" X 4 – 1/16" 10k, feed control line, CIL & DHPG cable through X-mass tree, tested all P. seals, Ok, applied pressure inside control line & rupture disk & flushed same, function tested TRSSV as open & close, Ok. Nipple up 4 – 1/16" X 4 – 1/16" X 4 – 1/16" X 4 – 1/16" 10k MSP solid block, pressure tested X-mass tree with 7500psi, Ok, removed 2-way BPV & installed one way, closed X-mass tree valves except bottom master valve.

Release rig 303 on 1396/10/16 at 06:00.

Work-Over#2

Spud Rig TDDC301 on 13/09/1398 at 10:30

after the end of well test & acidizing operation, it was reported that the annulus pressure has been raised up to maximum of 3960psi & maximum recorded wellhead pressure was 3750psi. After further investigations, it was decided to de-complete the well & re-complete the well with new completion string.

The well was de-complete, new completion string was ran. Several rans tried to retrieve plug & prong from No-Go landing nipple, No success. It was tried to unload the well above plug with water & diesel to retrieve the plug, No success. Left on top of Plug: Fish#1 Include: 2.15" GS Pulling (39.15cm) + 2" XOS (30cm) + 2.125" KNUCKLE Jt. (28cm) + 1.678" HYD. Jar (173cm) + 2" XOS (20cm) + 2.125" HSM (32cm). Totally Length: 3.32 mt.

RIH with coiled tubing & tried to retrieve the plug & prong with maximum over-pull.

Coiled tubing was parted about 21m above DIMPLE connector. Fish left in hole: 1st CT BHA (3.32m) + 2nd CT BHA (4.35m) + 21m of coiled tubing. Totally about 28.5m. TOF @3837.5m. R/D coiled tubing.

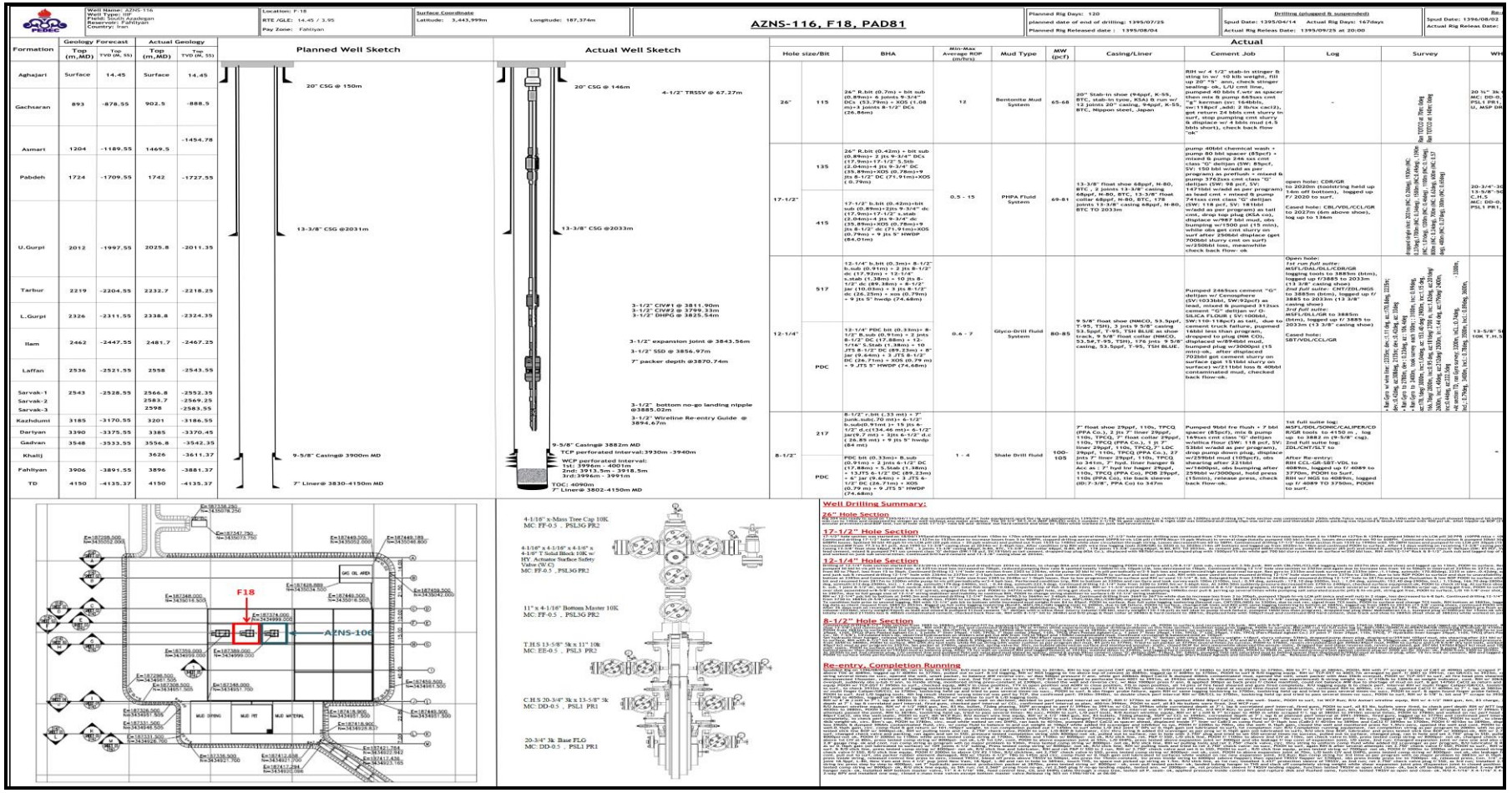
Fish #2: Include: 2" GS Pulling Hyd. Rel. (46cm) + 2 1/8" Hyd. Jar (207cm) + 2 1/8" Weight Bar (122cm) + MHA (60cm). Totally Length: 4.35 mt & 1.5 CTU Pipe (About 21.33 cm). Overall Fish Length: 29m. TOF: 3837.5m

Monitor WHP: 100^{psi}, annulus pressure, Zero. Perforated intervals are as following depths:

- 3930 – 3940^{m}
- 3996 – 4001^{m}
- 3913 – 3918.5^{m}

3996 – 3991^{m}Release rig 303 on 1400/03/04 at 24:00.

1.6 Well Montage



1.7 Casing & Liner Depth

Casing Spec.	20"	K – 55, 94#, BTC	0 – 146^{m}
	13 – 3/8"	N – 80, 68#, BTC	0 – 2033^{m}
	9 – 5/8"	T – 95, 53.5#, TSH Blue	0 – 3882^{m}
	7"	S – 110, 29#, TPCQ	3802 – 4149.5^{m}
	4 – ½"	T-95, 13.5#, Tenaris, 2 Jts blank + 7 Jts perforated	3968.9 – 4089.5^{m}

Table 4: AZNS-116 Casing & Liner Information

1.8 Well Schematic before Workover

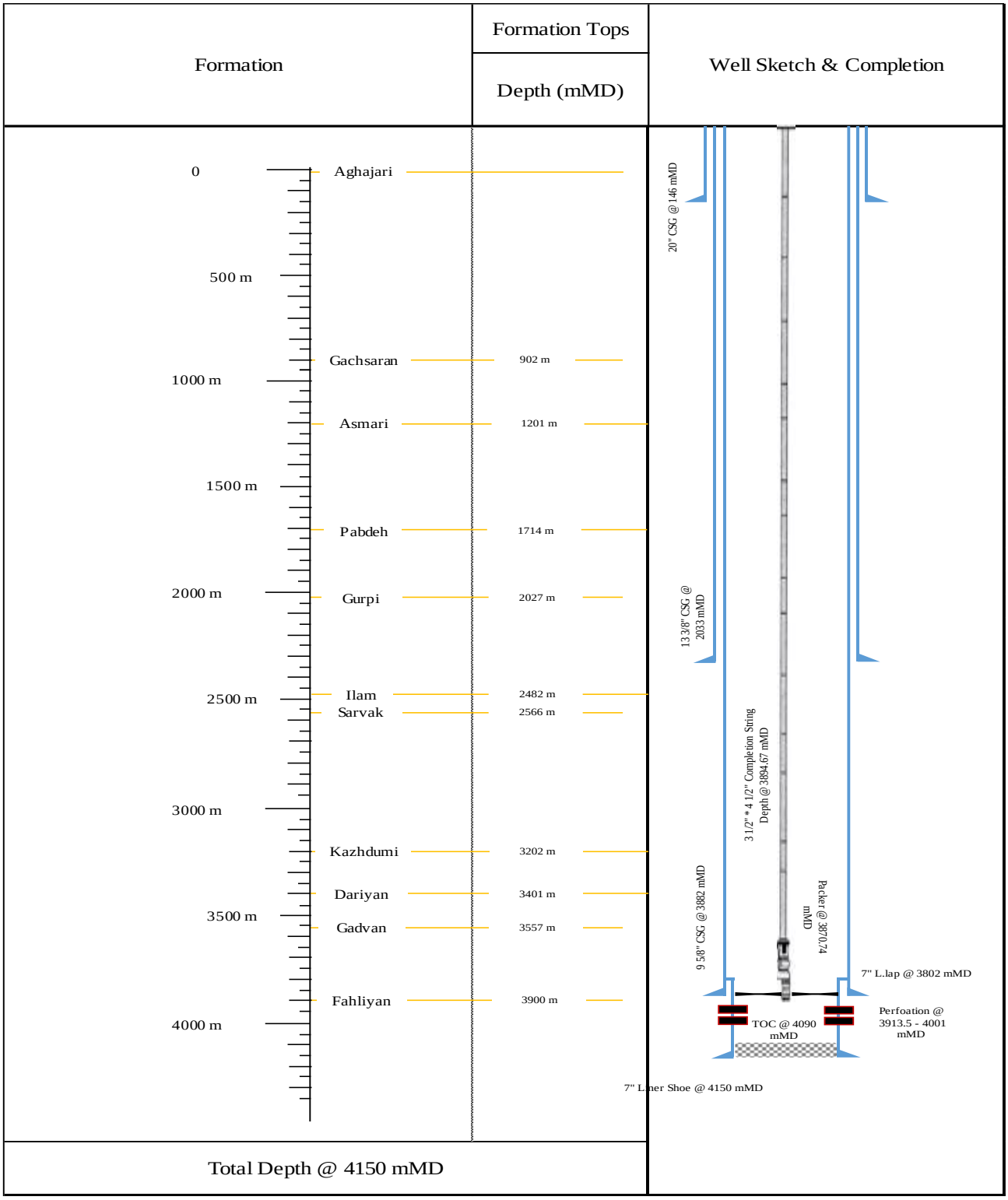


Figure 4: Well Schematic before Workover

1.9 Well Schematic after Workover

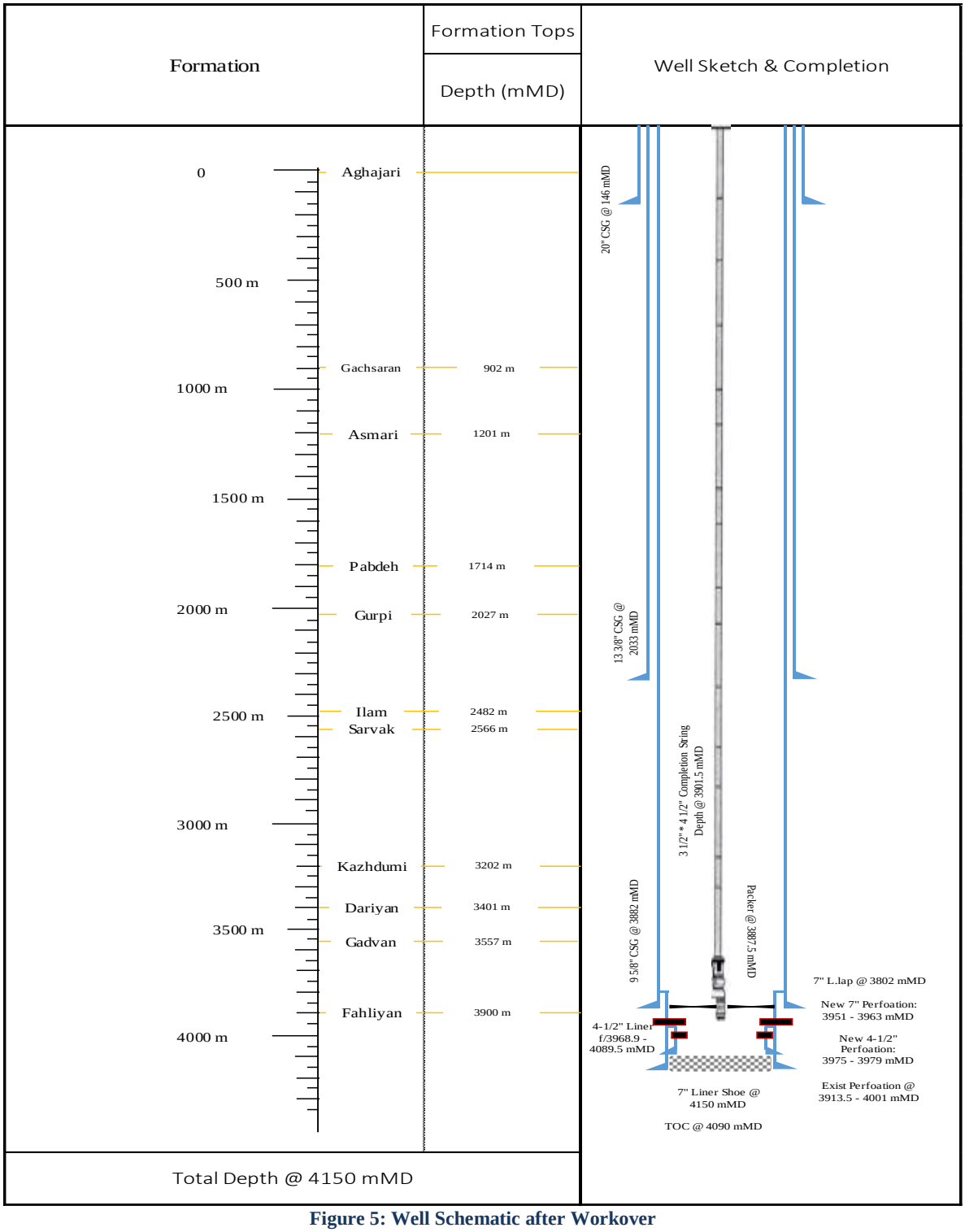


Figure 5: Well Schematic after Workover

2. Geological Information

2.1 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	- 4000	M,TVDss
Datum Pressure:	Fah – 4a = max 6800 Fah – 5a = max 7400	Psi
Pressure Gradient:	0.28	psi/ft.
Temperature:	132	°c
Temperature Gradient:	N/A	1.32 ft./100ft 0.73 C/100ft 2.4 C/100m
H_2S:	H_2S in Gas at Atm condition 3000 PPM	ppm
CO_2:	CO_2 in Gas at Atm Condition 50000 PPM	ppm
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 5: Reservoir Data

2.2 Prediction Pressure, Temperature & WOC

The formation pressure prediction, temperature & TTZ for Fahliyan sub layers has been estimated in AZNS – 116 based on the dynamic model as tabulated in table 6. The formation pressure prediction in block III is illustrated in table 7. The formation pressure prediction is the result of dynamic model.

Reservoir	Sub-layer	Mid of Sub-LayerDepth in AZNS-122 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure	Top Layer Depth (m)
Fahliyan	Fah-1a	-3888	252.9	1.32	9166	Dynamic Model	- 3885.4
	Fah-1b	-3898	253.3		9166	Dynamic Model	- 3890.7
	Fah-1c	-3922	254.4		8699	Dynamic Model	- 3905.6
	Fah-2a	-3943.5	255.3		8092	Dynamic Model	- 3938.5
	Fah-2b	-3954.7	255.8		8106	Pressure Gradient	- 3948.4
	Fah-3a	-3963	256.1		8696	Dynamic Model	- 3960.9
	Fah-3b	3971.5	256.5		8708.7	Pressure Gradient	- 3965.4
	Fah-4a	-3982.2	257.0		7710	Dynamic Model	- 3977.7
	Fah-4b	-4001.8	257.8		7729	Dynamic Model	- 3986.5
	Fah-5a	-4029.9	259.0		7817	Dynamic Model	- 4017
	Fah-5b	-4088.7	261.6		7906.8	Pressure Gradient	- 4042.7

Table 6: Predicted Pressure and Temperature and TTZ Contact Table

Formation	Interval	Pore Pressure	Collapse Pressure (Break-Out Pressure)	Fracture Pressure (Break-Down Pressure)
	m	*SG*	*SG*	*SG*
Aghajari	0 ~ 973	0.98 ~ 1.18	0.06 ~ 1.39	1.01 ~ 1.89
Gachsaran	973 ~ 1279	1.13 ~ 1.31	0.78 ~ 1.36	1.89 ~ 2.01
Asmari	1279 ~ 1473	1.14 ~ 1.26	0.6 ~ 1.28	1.43 ~ 1.79
Pabdeh	1473 ~ 2123	1.20 ~ 1.26	0.82 ~ 1.36	1.40 ~ 1.78
Gurpi	2123 ~ 2538	1.21 ~ 1.26	0.66 ~ 1.02	1.40 ~ 1.76
Ilam	2538 ~ 2612	1.18 ~ 1.24	0.20 ~ 0.9	1.14 ~ 1.72
Laffan	2612 ~ 2619	1.13 ~ 1.18	0.24 ~ 1.11	1.41 ~ 1.72
Sarvak	2619 ~ 3260	1.14 ~ 1.26	0.24 ~ 1.23	1.44 ~ 1.81
Kazhdumi	3260 ~ 3463	1.1 ~ 1.14	0.46 ~ 0.96	1.93 ~ 1.94
Dariyan	3463 ~ 3626	1.18 ~ 1.25	0.55 ~ 1.10	1.96 ~ 2.06
Gadvan	3626 ~ 3932	1.1 ~ 1.25	0.55 ~ 1.44	1.97 ~ 2.11
Fahliyan	3932 ~ TD	1.60 ~ 1.30	1.20~ 1.75	2.08 ~ 2.12

Table 7: Formation Pressure Prediction in Block III

2.3 Actual Formations Tops & Thicknesses

Table 8 illustrated the actual formation geological successions, top & thicknesses that penetrated in AZNS – 116 (all measured depths (MD) are based on RTE @14.45m).

Lithostratigraphic Unit	Formation op (m)				Vertical Thickness (m)	Difference (m) Forecast/VT	Casing Data	
	Forecast (m)	MD (sample, m)	MD (log, m)	Subsea (mc)			Size (in)	Depth (m)
Aghajari	Surface	Surface	Surface	+ 14.45	891.5	-	20"	146
Gachsaran	893	892	902	- 887.7	299.0	9	-	-
Asmari	1204	1182	1201	-1186.7	176.1	- 3	-	-
Pabdeh/Jahrum	1361	1375	1377	-1362.8	128.4	16	-	-
Jahrum	1493	1502	1506	-1491.2	208.3	13	-	-
Pabdeh	1724	1744	1714	-1699.5	311.8	- 10	-	-
U. Gurpi	2012	2026	2027	-2011.3	206.6	15	13 3/8	2033
Tarbur	2219	2230	2232	-2217.9	106	13	-	-
L. Gurpi	2326	2336	2339	-2323.9	143.5	13	-	-
Ilam	2462	2480	2482	-2467.4	76.5	20	-	-
Laffan	2536	2555	2558	-2543.9	7.1	22	-	-
Sarvak	2543	2564	2566	-2551	636.3	23	-	-
Kazhdumi	3185	3198	3202	-3187.3	199.3	17	-	-
Dariyan	3390	3396	3401	-3386.6	155.9	11	-	-
U. Gadvan	3548	3552	3557	-3542.5	70.7	9	-	-
Khalij Mbr	-	3622	3628	-3613.2	7.5	-	9 5/8	3882
L. Gadvan	-	3630	3635	-3620.7	260.7	-	-	-
Fahliyan	3906	3896	3900	- 3885.37	-	-6	-	-
TD	4150	4150	4150	- 4135.37	-	-	7	4150

Table 8: Actual Formation Tops & Thicknesses

3. Logging Program

The logging program has been tabulated in table 9, shall be performed by contractor in cased hole, section by section.

Hole Size (in)	Casing / Liner (in)	Cased hole wireline logging
12 – 1/4	9 – 5/8 Casing	Corrosion Log
8 – 1/2	7 liner	
8 – ½	7 liner	Perforation confirmation log (CCL/GR)
-	4 – ½ liner	Perforation confirmation log (CCL/GR)

Table 9: AZNS-116 Proposed Logging Plan

4. Program Outline

4.1 Contractors Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

De-Complete, Re-Complete Operation TBD of Well: AZNS-116 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	Trip	4150	48	2.00	2.0	2.2	52.8	2.20	2.20
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.	Wellhead	4150	48	2.00	4.0	2.2	52.8	2.20	4.4
Secure the well by close 3SV and X-Mass Tree Valves, Remove 7-1/16" XMT cap, R/U Slick line and perform pressure test,RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to top of fish .Note: If unable to pass to the bottom, follow contingency plan #1. Note:Chemical Wash May be Needed while RIH W/CT.	Trip	4150	40	1.67	5.7	1.8	44.0	1.83	6.2
R/U CT. RIH to top of fish@3837.5m.displace the well with heavy weitgh mud POOH and R/D CT.R/U Slick Line, Retrieve Protection Sleeve, Close 3SV, R/D S/L.	Trip	4150	72	3.00	8.7	3.3	79.2	3.30	9.5
Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOPs, open tie down screws,perform BOP pressure Test,if observe any leakage follow contingency plan#2 Tie down screw shall be provided by client.	Wellhead	4150	60	2.50	11.2	2.8	66.0	2.75	12.3
Retrieve TWCV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull/Rotation to un-latch the seal assembly. circulate to balance mud weight and flow check before pulling out. Note: if observe any damage on THS, follow contingency plan#2 Note: If unable to un-latch seal assembly, follow contingency plan #3.	Completion	4150	50	2.08	13.3	2.3	55.0	2.29	14.6
Pull out the completion string from anchor seal and lay down completion string precautionary due to probable trapped pressure below plug & Prong. While POOH monitor the well and annulus to make sure the well is safe and over balance.	Trip	4150	90	3.75	17.0	4.1	99.0	4.13	18.7
Perform BOP Test.	BOP	4150	6	0.25	17.3	0.3	6.6	0.28	19.0
RIH with 6" mill assembly, resume mill fish (CTU PIPE) & permanent packer, chase to 4-1/2" Perforated liner Lap @ 3968.9m (or where ever packer stopped), POOH. Note: If due to milling operation ctu pipe observe any deviation on well refer to contigency plan#4	Trip	4150	500	20.83	38.1	22.9	550.0	22.92	41.9
RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.Note: if no success to catch and recover packer, refer to contigenc plan #4	Trip	4150	150	6.25	44.3	6.9	165.0	6.88	48.8
RIH with 8-1/2" bit and clean out BHA to 7" liner lap @ 3802m.CBU. POOH to surface.	Trip	4150	48	2.00	46.3	2.2	52.8	2.20	51.0
Perform BOP Test.	BOP	4150	6	0.25	46.6	0.3	6.6	0.28	51.2
RIH with 5-7/8" to 6" Mill and clean out BHA to 4-1/2" Perforated liner Lap @ 3968.9m.CBU till hole clean. POOH to surface.	Trip	4150	60	2.50	49.1	2.8	66.0	2.75	54.0
RIH with 3-1/2" Mill and clean out BHA to bottom of 4-1/2" Perforated liner 4089.5m.CBU till hole clean. POOH to surface.	Trip	4150	120	5.00	54.1	5.5	132.0	5.50	59.5
RIH with 9-5/8" * 7" tandem scraper BHA, CBU & POOH to surface.	Trip	4150	36	1.50	55.6	1.7	39.6	1.65	61.1
RIH with wireline corrosion logging tools to bottom, log up to surface in two separate run (7" Liner & 9-5/8" Casing).	log	4150	40	1.67	57.3	1.83	44.0	1.8	63.0
Perform BOP Test.	BOP	4150	6	0.25	57.5	0.3	6.6	0.28	63.3
RIH with 5-7/8" mill and clean BHA and spot non solid fluid in front of perforation interval.	Trip	4150	36	1.50	59.0	1.7	39.6	1.65	64.9
Perform 4-1/2" liner perforation from 3975m to 3979m (Fah-3a)	perforation	4150	36.00	1.5	60.5	1.7	39.6	1.65	66.6
Perform 7" liner perforation from 3963 to 3951m (Fah-2a)	perforation	4150	36.00	1.5	62.0	1.7	39.6	1.65	68.2
P/U & M/U 5-7/8" Mill BHA, RIH to 4-1/2" liner lap @3968.9m while smoothing perforated intervals, CBU till shakers clean, POOH.	Trip	4150	36.00	1.5	63.5	1.7	39.6	1.65	69.9
P/U & M/U 3-1/2" Mill BHA, RIH to Btm while smoothing Perforated intervals, CBU shakers clean,POOH.	Trip	4150	60.00	2.5	66.0	2.8	66.0	2.75	72.6

Confirm 7" perforation interval with CCL logging tools, POOH to surface.	Log	4150	12.00	0.5	66.5	0.6	13.2	0.55	73.2
Confirm 4-1/2" perforation interval with CCL logging tools, POOH to surface.	Log	4150	12.00	0.5	67.0	0.6	13.2	0.55	73.7
RIH with tandem 7" scrapers to top of Perforation intervals, Displace the well w/ Completion Fluid, POOH while L/D all DPs, DCs & Tools.	Trip	4150	72.00	3.0	70.0	3.3	79.2	3.30	77.0
Run 10K completion string as per program. NOTE: Annulus pressure test during completion operation is related to corrosion log result.	Completion	4150	125	5.21	75.2	5.7	137.5	5.73	82.7
Set BPV inside TBH, N/D BOP's & install new 10K XMT.	Completion	4150	48	2.00	77.2	2.2	52.8	2.20	84.9
Retrive BPV, install TWCV and Pressure test the XMT.	Completion	4150	24	1.00	78.2	1.1	26.4	1.10	86.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion	4150	10	0.42	78.6	0.5	11.0	0.46	86.5
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with C/T & perform acid wash as per program, POOH to surface.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
RIH with S/L, retrieve protection sleeve.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion	4150	Rig Less	0.00	78.6	0.0	0.0	0.00	86.5
Total Duration of Work-over:				*78.6*	*78.6*			*86.5*	*86.5*
NOTE:									
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4250	10	0.42	79.0	0.5	11.0	0.46	86.9
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4250	44.00	1.83	80.9	2.0	48.4	2.02	89.0
Total Duration of AZNS-116 Work-over Well:		4250			80.9				89.0
Contingency Plans TBD of Well: AZNS-116 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed									
Contingency plan #1: if observe any obstruction while each run w/ Gauge Cutter RIH w/CT and wash from obstruction depth to top of fish with Zylyne.	Trip	4250	96	4.00	4.0	4.4	105.6	4.40	4.4
Contingency plan #2: if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG). Note:Proper Equipment shall be provided by client and install by contractor.	Completion	4250	200.0	8.3	12.3	9.2	220.0	9.2	13.6
Contingency plan #3: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.	Completion	4250	320.0	13.3	25.7	14.7	352.0	14.7	28.2
Contingency plan #4: perform fishing and catch the completion packer with propper fishing metod	Completion	4250	200.0	8.3	34.0	9.2	220.0	9.2	37.4
Contingency plan #5: as per client request perform dry test and reated to result run and set SN. Packer.	Completion	4250	170.0	7.1	41.1	7.8	187.0	7.8	45.2
Total Duration of AZNS-116 Work-over Well:		4250			19.4				45.2

5. Sequence of Operation

5.1 Main Objective

Based on fishing operation results the decision has been made to de-complete the well, perforate new intervals, change completion string & re-complete the well.

5.2 Main plan

1. Held PJSM. Spud the Rig. Stand back required 5" & 3-1/2" DPs, 5" HWDPs & 6-1/2" DCs.
2. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval & recheck the well status.
3. Secure the well by close 3SV & X-Mass Tree valves, remove 7 – 1/16" XMT cap, R/U slickline & perform pressure test, RIH w/ gauge cutter to top of 3SV, RIH & set protection sleeve, RIH w/ gauge cutter to top of fish.

Note: If unable to pass to the bottom, follow contingency plan #1.

Note: Chemical wash may be needed while RIH w/ CT.

4. R/U CT, RIH to top of fish @3837.5^{m}, displace the well with heavy weight mud, POOH & R/D CT. R/U slickline, retrieve protection sleeve, close 3SV, R/D slickline.
5. Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOPs, open tie down screws, perform BOP pressure Test.

Note: If observe any leakages, follow contingency plan #2.

Note: Tie down screw shall be provided by client.

6. Retrieve TWCV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull / rotation to un-latch the seal assembly. Circulate to balance mud weight & flow check before pulling out.

Note: If observe any damages on THS, follow contingency plan #2.

Note: If unable to un-latch seal assembly, follow contingency plan #3.

7. Pull out the completion string from anchor seal & lay down completion string precautionary due to probable trapped pressure below plug & prong. While POOH, monitor the well & annulus to make sure the well is safe & over balance.
8. Perform BOP test.
9. RIH with 6" mill assembly, resume mill fish (CTU pipe) & permanent packer, chase to 4-1/2" perforated liner lap @3968.9^{m} (or wherever packer stopped), POOH to surface.

Note: If due to milling operation CTU pipe observe any deviation on well, refer to contingency plan #4.

10. RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.

Note: If no success to catch & recover the packer, refer to contingency plan #4.

11. RIH with 8 – 1/2" bit & clean out BHA to 7" liner lap @3802^{m}. CBU. POOH to surface.
12. Perform BOP test.
13. RIH with 5 – 7/8" to 6" mill & clean out BHA to 4-1/2" perforated liner lap @3968.9^{m}, CBU till hole clean, POOH to surface.
14. RIH with 3-1/2" mill & clean out BHA to bottom of 4-1/2" perforated liner @4089.5^{m}, CBU till hole clean, POOH to surface.

Note: This Operation has a high risk of String stuck (Metal to Metal) due to low clearance.

15. RIH with 9 – 5/8" * 7" tandem scraper BHA, CBU & POOH to surface.

Note: Choke manifold & choke line must be lined up for **HARD SHUT-IN**

16. RIH with wireline corrosion logging tools to bottom, log up to surface in two separate runs (7" Liner & 9 – 5/8" Casing).
17. Perform BOP test.
18. RIH with 5-7/8" mill & clean out BHA & spot non – solid fluid in front of perforation intervals.
19. Perform 4-1/2" liner perforation from 3975^{m} to 3979^{m} (Fah-3a)

20. Perform 7" liner perforation from 3963^{m} to 3951^{m} (Fah-2a).
21. P/U & M/U 5-7/8" mill BHA, RIH to 4-1/2" liner lap @3968.9^{m} while smoothing perforated interval, CBU till shakers clean, POOH.
22. P/U & M/U 3-1/2" mill BHA, RIH to bottom while smoothing perforated interval, CBU till shakers clean, POOH.
23. Confirm 7" perforation interval with CCL logging tools.
24. Confirm 4-1/2" perforation interval with CCL logging tools.
25. RIH with 7" tandem scraper to top of perforation intervals, displace the well with completion fluid, POOH while L/D all DPs, DCs & tools.
26. Run 10k completion string as per program.

Note: Annulus pressure test during completion operation is related to corrosion log result.

27. Set BPV inside TBH, N/D BOP's & install new 10k XMT.
28. Retrieve BPV, install TWCV & pressure test the XMT.
29. Secure the well by set BPV in TBG hanger, close all X-mas Tree valves except BMV & release the Rig.
30. Retrieve TWCV, R/U slickline, set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.
31. RIH with CT to bottom, displace the well with diesel or nitrogen & observe well, POOH.
32. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.
33. Flow the well as pre – acid clean up till acceptable BSW & reading stable flow rate on 2 different choke sizes for 72hrs.
34. RIH with CT & perform acid wash as per program, POOH to surface.
35. Flow the well as post – acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke sizes for minimum 96hrs.

Note: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

Note: Acidizing & post – acid clean-up maybe performs as rig less job upon contractor request & client approval, but in case of any failure in well integrity, the contractor should provide drilling rig & required services in own responsibility & fix the problem.

36. RIH with SL, set protection sleeve, POOH & R/D SL.
37. RIH with CT to bottom & displace the well with diesel, POOH.
38. RIH with SL, install plug & prong into NO GO landing nipple, POOH.
39. RIH with SL, retrieve protection sleeve.
40. Secure the well by set BPV in TBG hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS – 116 Work-over:

Based on operation situation in each step, the following contingency plans might be followed:

Contingency plan #1: If observe any obstruction while each run w/ gauge cutter, RIH w/ CT & wash from obstruction depth to fish with Zylyne.

Contingency plan #2: If observe any leakage & or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG).
Note: Proper equipment shall be provided by client & install by contractor.

Contingency plan #3: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.

Contingency plan #4: Perform fishing & catch the completion packer with proper fishing method.

Contingency plan #5: As per client request perform dry test & related to results, run & set SN packer.

5.3 Perforation plan

Perforation gun & charges be provided according to below table. CCL – GR shall be provided in requested sizes. Guns can be run on wireline or TCP be performed depending on total perforation length.

Gun OD (in)	Shot density (SPF)	Phase angle	API Penetration Depth (in)	Explosive type	Under-balance pressure(psi)

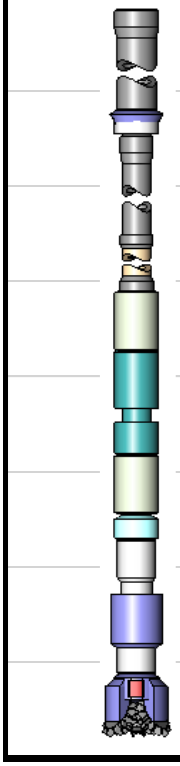
4.5	5	72°	≥40	HMX	NA
2 7/8 or 2 1/8	6 6	45° 45°	≥25 ≥21	HMX	NA

Table 10 - Proposed Perforation Gun

6. BHA Program

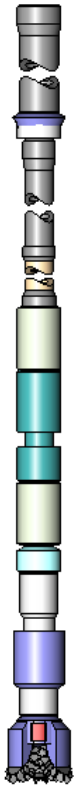
6.1 8-1/2" BHA

BIT SIZE (inch)	8-1/2"	BHA Configuration to C/Out Inside 9-5/8" Casing				Collar	OD
Depth In/Out (m)						DC-1	6-1/2"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0.784						
Weight in Air (lb)	24,624						
Bouyed Weight (lb)	19,297						
Bouyed Weight Below Jar (lb)	14,267						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27.42	182.71
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	1
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27.42	155.29
	6-1/2" Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	8	127.87
	6-1/2" Drill Collar	13	6-1/2	2-13/16	4-1/2" Reg. P x B	118.82	119.87
	6-1/2" Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x B	0.75	1.05
	8-1/2" MT Bit	1	8-1/2"	-	3-1/2" Reg. P x B	0.3	0.3

6.2 5-7/8" or 6" BHA

Mill SIZE (inch)	5-7/8" or 6"	BHA Configuration to C/Out Inside 7" Liner				Collar	OD
Depth In/Out (m)						DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0.784						
Weight in Air (lb)	29,642						
Bouyed Weight (lb)	23,229						
Bouyed Weight Below Jar (lb)	16,462						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

7. Mud & Cement Program

Note: Detailed mud & cement program, will be issued in due time as per client approved template.

7.1 Workover Hole Section

Workover section	7" Liner
From: 3802m	To: 4150m
Kill the well, POOH of Completion String, Complete the hole with Completion 10k & Completion Fluid	

Overview

Drilling Fluid System	Kill Mud and Completion Fluid
Key products	Workover Fluid , Completion Fluid, Kill Mud
Water type	Drill Water
Solid Control	Shale Shaker
Potential problems	Down Hole losses , gas/oil influx, H_2S

Interval Mud Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
0 – 4150	103 – 106	30 – 55	8 – 22	NC	9 – 10.5	45 – 55

Interval Completion Fluid Properties						
Depth interval (m)	Mw (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface ~ 3950	104	ALAP	N/A	NC	6 – 7.5	27 – 30

Mud Discussion & Mixing Procedure

7.1..1 Mud type

Kill mud 103 – 106^{pcf} will be used to kill the well & well will be displaced by Completion fluid of $CaBr_2$ 104^{pcf}. Attention must be paid that there is Calcium Bromide Brine inside the well & all of them must be stored in separated mud tank by the time of displacing the well by kill mud. (As what the hole dictates, mud properties may be varied by the time).

7.1..2 Drilling Fluid Formulation

Kill Mud/Work over Mud formulation		
Component	**Conc. (ppb)**	**Component Description**
Drill water	0.5766 bbl/bbl	Base Fluid
Caustic Soda	0.3	PH Modifier (lime can be used instead)
Soda Ash	0.3	Hardness Remover and
Sodium Chloride	63	Non-Solid weighting Agent
Xanthan-Gum	0.2	Viscosifier (In case of need and based on situation can be used)
Calcium Carbonate	210	Weight Agent
Hematite	As needed	Weight Agent

*Above calculation is based on mud weight 103 – 106^{pcf}

*Lime can be substituted for caustic soda

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Calcium Bromide (14.2^{ppg})	0.984 bbl/bbl	Base Fluid
Corrosion Inhibitor	1% Vol	Anti-Corrosion

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
H_2S Scavenger Liquid	0.5% Vol	H_2S Removal/ Anti Corrosion
Oxygen Scavenger CA	0.5 ppb	Oxygen Removal/Anti Corrosion

The performance requirements designed for the completion fluid system are as follows:

- Recovered permeability: ≥ 90 ٪
- Corrosion rate ＜0.076 mm/year
- Diameter of solid particle ≤ 2μm
- For fluid pumping in the well NTU<30
- For fluid returning from the well: NTU <150
- Acid Soluble/ Removable with Solvent

7.1..3 Drilling Fluid Formulation

Kill Mud Mixing Procedure:

1. Fill tanks with calculated amount of Drill Water 0.576 bbl/bbl.
2. Treat Drill Water with Caustic soda & Soda Ash by 0.3pbb.
3. Mix NaCl at 66 ppb.
4. Add Xanthan-Gum by 0.2pbb.
5. Add $CaCO_3$-F (210ppb) to reach Mw of ±95^{pcf}.
6. Add Hematite (as needed) to reach Mw of ±103^{pcf}.

   ↳ In case of foaming use Alcohol-Based Defoamer.

Note: Use Hi Vis pill (existing drilling fluid with viscosity ±70) to clean hole if necessary. Calculation is based on per bbl of mud.

Completion Fluid Mixing Procedure:

1. Make sure that all pits & lines involved with completion fluid are thoroughly cleaned, flushed & free of any fluid and solid.
   - Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed & cleaned for accommodating the brine of next ready pit, & so on to the last pit.
   - Depends on the rig facilities & structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.
2. Transfer the content of $CaBr_2$ drums into designated pits.
   - Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.
3. Mix Corrosion Inhibitor into pits at concentration of 3.5^{pbb} (1% by volume).
4. Mix H_2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators & mixing pumps, & leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside (mostly insoluble Lime) settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.
   - Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.
8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the & same above method.

Note: all $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.
Based on well readings & the operation, mud weight may be varied.

7.1..4 Downhole Losses

As perforation had been done in well AZN#116 & there is exposure to reservoir, risk of any probable formation losses must be taken into consideration.
Losses will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses

are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of $CaCO_3$ 50 micron & 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

Seepage losses Less than 10 bbl/hr.

No specific action will be taken, but close monitoring of the probable loss rate is made.

Losses from 10 – 30 bbl/hr.

- Load $CaCO_3$-M (150 – 300 micron) in system up to 5^{ppb}.

***If loss not cured & even tends to increase, then:**

- Pump 40 – 60^{bbl} of pill made by 7 – 10^{ppb} $CaCO_3$ 150 – 300^{μ} & 7 – 10^{ppb} $CaCO_3$ 600^{μ}

Losses from 30 – 50 bbl/hr.

- Load $CaCO_3$-M (150 – 300 micron) in system up to 5^{ppb}.
- Since some of the $CaCO_3$-M will be discarded over the shakers, $CaCO_3$-M to compensate the separated amount on the shakers.

***If loss not cured & even tends to increase, then:**

- Pump 40 – 60^{bbl} of pill made by
- 10 – 20^{ppb} $CaCO_3$ 200 – 300^{μ}
- 10 – 20^{ppb} $CaCO_3$ 600^{μ}
- 10^{pbb} Oyster Shell-F
- 10^{pbb} Oyster Shell-M.

Losses from 50 – 100 bbl/hr.

- Load $CaCO_3$-M (150 – 300 micron) in system up to 5^{ppb}.
- Load $CaCO_3$-C (600 – 100 micron) in system up to 5^{ppb}.
- Since some of the $CaCO_3$-M & C will be discarded over the shakers, add $CaCO_3$-M & C to compensate the separated amount on the shakers.

***If loss not cured & even tends to increase, then:**

- Pump 40 – 60^{bbl} of the following pill:

- 10 – 20^{ppb} $CaCO_3$ 200 – 300 micron.
- 10 – 2 ^{ppb} $CaCO_3$ 600 micron.
- 7 – 10^{ppb} of each $CaCO_3$ 1000 micron & $CaCO_3$ 2000 micron.

- 5 – 10[pbb] Oyster Shell -F
- 5 – 10[pbb] Oyster Shell -M

Losses more than 100 bbl/hr.

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F & M & C should be pumped and spotted & gently squeezed over the loss zone.

Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes & monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP & squeeze 10[bbl] LCM pill into formation without exceeding 200[psi].
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix & spot another 50[bbl] of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP & squeeze 20[bbl] of LCM pill into the formation without exceeding 200[psi].
6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string back to bottom.

Notes:

- LCM pills must be thoroughly homogenised & viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe.
- Remove filter screens from the drill pipe (if there is any) & if necessary, remove strainers from the mud pump suction lines.

7.1..5 Engineering Comments

Keep watching and control on well to record loss.
All $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.

7.1..6 Drilling Issues

There is no drilling objective in AZN-116 workover well; mud should be compatible with reservoir and work over operation.
Solid Control equipment

- Use shale shaker to remove debris and LCM particle in case of use.
- Personal Safety

7.1..7 Mud Type

Inhibited $CaBr_2$ brine (Calcium Bromide Brine) will be applied as completion fluid. And simple salt saturated kill mud to kill the well & clean out inside 9 5/8", 7" Liner.

Displacement Design

The combined effect of scrapers being run in the hole, the high flow rate will help to remove/disperse general debris.
At least 150^{m} of pill/spacer coverage or 3min. contact time at the highest pump rates should be considered.
Saturated $CaCl_2$ brine can be applied as spacer/Hi-Vis pill while displacing.
Pit levels should be monitor throughout the displacement operation to catch & prevent an influx entering the well contributing to an incident.

No.	Description	Volume [bbl.]	Comment
1	Viscous pill/spacer	50	Clean pill
2	Displace to inhibited $CaBr_2$ Brine at 104^{pcf} (Completion Fluid)		
3	Continue pumping Completion Fluid until cleanliness specification has been met.		

All $CaBr_2$ retrieved from well must be stored in separated tanks. All $CaBr_2$ left on surface, even dead volume of tanks, must be transferred back to drums at end of well. Not even a drop of $CaBr_2$ is not allowed to be wasted or dump in whole operation duration.
It is recommended to use desander tank as active pit of $CaBr_2$ to minimize dead volume & return path.

7.1..8 Volume accounting & estimated required material

Work over /Kill Mud Fluid				AZADEGAN #116	
Fluid Density (pcf)	103 – 106		Fluid System	Water base Work Over Fluid /Salt Saturated	
Hole Size (inch)	Completion Phase				
Casing Depth (m)	3882		Solids Control Equip Eff.%		
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)		4150
Liner Depth (m)	4150		Liner Lap (m)		3802

Liner OD / ID (inch)	7	6.276	Liner Length (m)	348

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	350
Cased Hole Volue [Capacity of 3882^{m} 9⅝" Casing + Capacity of 348^{m} 7" Liner]	960
Total New Volume Required (bbl)	**1310**

Estimated Chemical Consumption and Cost based on 103-106pcf MW

PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Drill Water	bbl	N/A	0.576 bbl/bbl	755		755	N/A
Caustic Soda	25 Kg	N/A	0.3	7	25	32	N/A
Soda Ash	50 Kg	N/A	0.3	4	16	20	N/A
Sodium Chloride	1 MT	N/A	63	37	18	55	N/A
Xanthan-Gum	25 Kg	N/A	0.2	5	20	25	N/A
Calcium Carbonate	1 MT	N/A	325	194	30	224	N/A

*Above calculation are based on normal hole condition

*Above Calculated is Based on 103 – 106^{pcf}

Completion Fluid			AZADEGAN #116	
Fluid Density (pcf)	104		Fluid System	$CaBr_2$ Brine
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3882		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4150
Liner Depth (m)	4150		Liner Lap (m)	3802
Liner OD / ID (inch)	7	6.276	Liner Length (m)	348

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volume	960

[Capacity of 3882$_m$ 9⅝" Casing + Capacity of 348^m 7" Liner]	
Total New Volume Required (bbl)	**1210**

Estimated Chemical Consumption & Cost based on 74 – 75^{pcf} Mw							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.984bbl/bbl	910	62	972	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	9	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	11	6	17	N/A

*Above calculation are based on normal hole condition
*Above Calculated is based on **104^{pcf}**

7.1..9 Estimated contingency material

During all the phase keep a minimum safety stock of below items:

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe – Freeing Agent	55 Gal DM	4	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	40	Cement Contamination
Citric Acid	25 Kg Sx	40	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160 each	Loss circulation material

Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant	55Gal	8	Biodegradable lubricator

7.1..10 Material requirements for each well

Item	QTY	Unit	Description	Function
1	972	55 Gal Dr	Calcium Bromide (14.2 ppg)	$CaBr_2$ brine
2	55	1 MT	Salt	Weighting Agent
3	224	1 MT	Calcium Carbonate 75 Mic	Weighting Agent
4	32	25 Kg Sx	Caustic Soda	PH Control
5	20	50 Kg Sx	Soda Ash	Hardness reducer
6	25	25 Kg Sx	Xanthan-Gum	Viscosifier
9	80	25 Kg Sx	Lime	PH Control
10	160	25 Kg Sx	Shell Fish	LCM
11	160	25 KG Sx	Limestone Chips	LCM
12	10	55 Gal Dr	Corrosion Inhibitor	Completion Fluid Corrosion Inhibitor
13	8	55 Gal Dr	H2S SCAV	Completion Fluid H_2S SCAV
14	17	25 Kg Sx	OX SCAV	Completion Fluid OX SCAV
15	1	55 Gal Dr	BIO-CIDE	Anti-bacterial
16	12	55 Gal Dr	PH BUFFER	PH Stabilizer

7.2 Cement Program

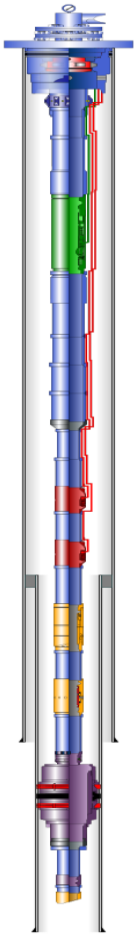
Not applicable.

8. Completion

8.1 Proposed Completion string

AZNS-116 Completion Sketch

Contractor	MSA	Liner & Tubing Data					Client	PPL/PEDEC
Field	AZADEGAN	Description	7"Liner	3 1/2" Tubing	4 1/2" Tubing	Tubing	7'' Tie back Liner Lap:	-
Well Number:	AZNS-116	OD[in]	7.00	3.50	4.50		7'' Liner lap:	3802
Rig :	OEOC-201	Nominal weight [lb/ft]	29.00	10.20	18.90		5' Liner Shoe:	-
RTE-GLE	13.45	Grade	110S	L-80	L-80		5" Liner Shoe	-
RTE-THF	3.95	Thread	TPCQ	HSM3	HSM3		Total Depth:	4150
Rig Spudded Date :	TBA	Inside diameter [in]	-	2.92	3.640		X-mass Tree:	MSP-T TYPE
End of Operation	TBA	API Drift Diameter	-	2.80	3.515		Size & Rating:	4 1/16"- 10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				11.69	0.000
	2	TBG HANGER 11" * 4 1/2'' ACME Box up * 4 1/2'' TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type ''H'' Two Way	11	3.68	0.6	12.29	11.690
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	6.1	3.64	1.01	13.300	12.290
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.64	60.940	13.300
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	62.560	60.940
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.72	64.280	62.560
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type,	7.5	3.437	1.78	66.060	64.280
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.11	67.170	66.060
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	68.790	67.170
	10	392 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3731.84	3800.630	68.790
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.02	3801.650	3800.630
	12	5 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	19.04	3820.690	3801.650
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3822.510	3820.690
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3823.240	3822.510
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3825.060	3823.240
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3834.560	3825.060
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3836.380	3834.560
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3837.110	3836.380
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	3838.910	3837.110
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3848.410	3838.910
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3850.230	3848.410
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.135	2.922	5.755	3855.985	3850.230
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3857.807	3855.985
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3867.307	3857.807
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3868.932	3867.307
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	3870.122	3868.932
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	3871.822	3870.122
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	3881.092	3871.822
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3882.817	3881.092
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4/00	2.922	0.23	3883.047	3882.817
	31	7" 29 PPF CASING HYDRAULIC SET RERMANENT PRODUCTION PACKER TO BE RUN IN 7.00" 29 PPF CASING	5//910	2.922	4.44	3887.487	3883.047
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	3889.309	3887.487
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3898.809	3889.309
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3900.634	3898.809
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.562" TYPE XN (NO-GO ID: 2.442'')	4.5	2.562	0.37	3901.004	3900.634
	38	RE-ENTERYY MULE SHOE	4.25	2.922	0.37	3901.374	3901.004
		END OF COMPLETION STRING					3901.374

Tubing Retrievable Saftey Valve	66.060	64.280
Traveling Joint	3855.985	3850.230
CIV 1	3823.240	3822.510
CIV 2	3837.110	3836.380
SSD	3870.122	3868.932
7'' HYDRAULIC SET PERMANENT PRODUCTION PACKER	3887.487	3883.047
NO-GO Landing Nipple	3901.004	3900.634
Entry Guide	3901.374	3901.004

Figure 6: Proposed Completion Sketch

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree



Figure 7: Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20 – 3/4" 3k	13 – 5/8" 5k * 20 – 3/4" 3k	11" 5k * 13 – 5/8" 5k	5k
Gadvan / Fahliyan	20 – 3/4" 3k	13 – 5/8" 5k * 20 – 3/4" 3k	11" 10k * 13 – 5/8" 5k	10k
Water disposal wells	20 – 3/4" 3k	13 – 5/8" 5k * 20 – 3/4" 3k	11" 5k * 13 – 5/8" 5k	5k

Item	Specification
Casing Head Housing assembly	MC: DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC: DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 11: AZNS Wellhead Information

- **Wellhead Specification**

NOTE: Wellhead and X-Mass Tree already installed by client.

20 – 3/4" * 13 – 5/8" * 11" * 4 – 1/16" 10,000^{psi} WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94$^{lb/ft}$) OD casing * 20 – 3/4" 3,000^{psi} API flange **has hold down bolts for wear bushing,** with 2 * 2 – 1/16" 3,000^{psi} studded side outlets, test ports & seals. One side outlet has **one** 2 – 1/16" 3,000^{psi} manual gate valve, 2" LP threaded companion flange * 2" bull plug threaded for ½" NPT & fitted with 0 – 3,000^{psi} gauge & needle valve &, the other side to have 2 – 1/16" 3,000^{psi} * 2" LP threaded flange & 2" bull plug. Casing slip & seal assembly self – energizing to hang 13 – 3/8" (68$^{lb/ft}$) OD casing. This section installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630^{mm}** **Material:** Forged alloy steel 4130 **Automatic casing slip material:** Low alloy steel 4130
Section B	**Casing head spool** with 13 – 5/8" 5,000^{psi} top flange * 20 – 3/4" 3,000^{psi} bottom flange has **hold down bolts for wear bushing** & bottom preparation to seal on 13 – 3/8" OD casing with test ports to test annular space & casing seals. One side outlet has **one** 2 – 1/16" 5,000^{psi} manual gate valve, 2" LP threaded companion flange & 2" bull plug on one side &, the other side has **one** 2 – 1/16" 5,000^{psi} manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT & fitted with 0 – 5,000^{psi} gauge & needle valve. Casing slip & seal assembly self – energizing to hang 9 – 5/8" (47$^{lb/ft}$) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665^{mm}** **Material:** Forged low alloy steel 4130 **13 – 5/8" * 9 – 5/8" automatic casing slip material:** Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000^{psi} top flange * 13 – 5/8" 5,000^{psi} bottom flange has **hold down bolts for wear bushing** & bottom preparation to seal on 9 – 5/8" OD casing with test ports to test annular space & casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 – 1/16" 10,000^{psi} manual gate valve, 2"ACME 10k female thread companion flange, 2" bull plug threaded for ½"NPT & fitted with 0 – 10,000^{psi} gauge & needle valve on one side &, the other side to have the first 2 – 1/16" 10,000^{psi} manual gate valve, 2 – 1/16" 10,000^{psi} instrument flange installed with one ½"NPT & fitted with 0 – 10,000^{psi} gauge & needle valve, & the second 2 – 1/16" 10,000^{psi} manual gate valve, 2" ACME 10k female thread companion flange & 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8 – 1/2" BHA. The 11" * 4 – 1/2" **tubing hanger** has 4 – 1/2" 18.9^{ppf} premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for 3/8" chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck & two S-SEAL to pack off the body of hanger, & tonging preparation on the lower neck. Tubing hanger shall have factory – torqued double pin tubing pup joint 4 – 1/2" 18.9^{ppf}, L80, New Vam approximate length 1600^{mm} to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850^{mm}** **Material of body:** Forged low alloy steel 410SS

	G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000^{psi} studded flange * 4 – 1/16" 10,000^{psi} full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) & with test ports to test annular space & seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG & two channels for 3/8" chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 – 1/16" 10,000^{psi} bottom flange for tubing adaptor. 4 – 1/16" 10,000^{psi} lower manual gate valve, 4 – 1/16" 10,000^{psi} upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 – 1/16" 10,000^{psi} manual swab valve. With two 4 – 1/16" 10,000^{psi} studded outside for wing valves. With one 4 – 1/16" 10,000^{psi} studded top flange for tree cap. (3) one wing valve 4 – 1/16" 10,000^{psi} full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0 – 10,000^{psi} pressure gauge & needle valve installed & one 4 – 1/16"10,000^{psi} Cage type positive choke with 90 degree outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch & with welding flange prepared for 4" pipe (specification to be confirmed). (4) The other wing with one 4 – 1/16" 10,000^{psi} full-bore manual gate valve & one companion flange w/2" ACME 10k female thread. (5) Quick union type Tree Cap above, with 4 – 1/16" 10,000^{psi} flange at bottom, ACME thread union on top, including 1/2" NPT port for needle valve & pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800^{mm}**. **Material of body:** 410SS **G. valves, material of seats/gates & stem:** Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 12: AZNS Wellhead Specification

9.2 BOP

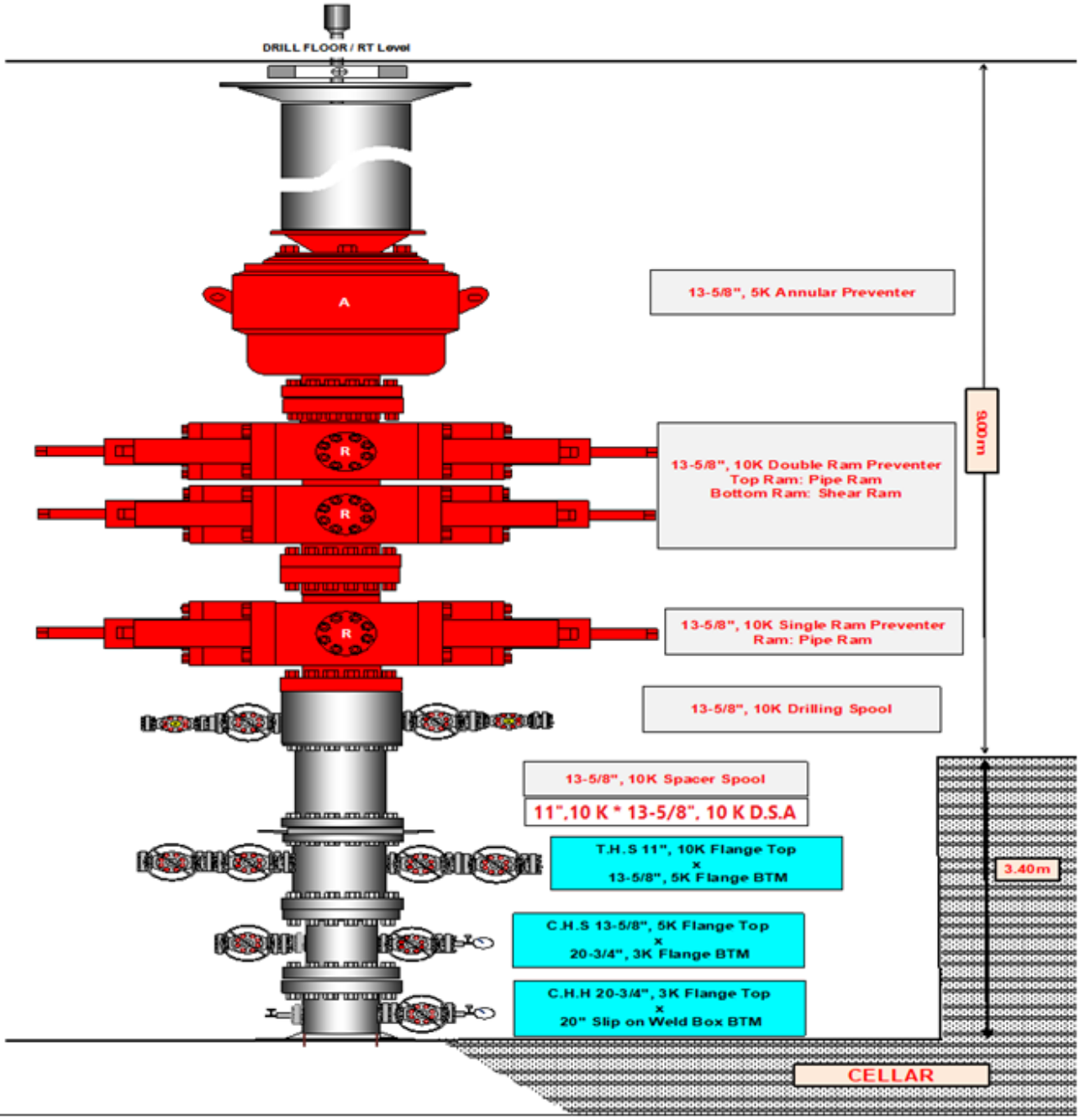


Figure 8: BOP Stack

10.

10. Waste Management

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.

When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.

Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:

Each time before POOH after drilling oil and gas formations.

Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil and gas formation.

POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;

Hard shut-in method is recommended for oil well;

Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if

special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.
- Logging Results
- BOP Test Report
- Service Company Report

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	8 – 1/2" Milled Tooth Bit	2	Ea.	MSA
2	5 – 7/8" or 6" Mill	8	Ea.	MSA
3	5 – 7/8" or 6" Rock bit	2	Ea.	MSA
4	7" Liner hanger with SN Packer	1	Set	MSA
5	4 – 1/2" * 3 – 1/2" Completion String 10k	1	set	MSA
6	9 – 5/8" & 7" Scraper	1+1	ea.	OEOC
7	5" DP, HWDP, 8 – 1/2" DC,	as required		OEOC
8	Well Head Services	as required		Service Co.
9	Surface Well Test Services	as required		Service Co.
10	Completion Services	as required		Service Co.
11	Slick line Services	as required		Service Co.

12	Casing Running Services	as required		Service Co.
13	Wireline Logging	as required		Service Co.
14	Dry Test, DST, RTTS & Bridge plug	as required		Service Co.
15	Cementing Services	as required		Service Co.
16	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
17	H2S Services	as required		Service Co.
18	Mud Logging Services	as required		Service Co.
19	Rig Services	as required		Service Co.
20	Drilling & Completion Fluid Services	as required		Service Co.
Note: Quantity & Spec to be conformed with contract's SOW. Note: In case of any shortage or discrepancy, the contract's SOW will be reference.				

Table 13: List of Requirements

15. Revision Corrections

Drilling Engineer							
1	**Well Information:** • مدت زمان 89 روز در نظر گرفته شده اصلاح گردد	7	CM	• با توجه به اینکه در شرح کار و تاریخچه ارسالی کارفرما اشاره ای به وجود آستری 4-1/2 اینچ در چاه نشده بود و اخیرا به شرح کار اضافه گردید مدت زمان تعمیر چاه با توجه به عملیات های مربوط به آستری مذکور افزایش یافت.			
2	**Time Breakdown:** • مدت زمان درنظر گرفته شده برای عملیات تمیز سازی داخل آستری 4 1/2 مجدداً بررسی گردد.	43-44	CM	• بر اساس گزارشات روزانه و تاریخچه چاه گردش با گل سنگین بالای آستری 4-1/2 اینچ به مدت ده ماه بدون تمیز سازی درون آستری صورت پذیرفته است لذا تمیز سازی درون آستری زمان بر خواهد بود، لازم به توضیح است در صورت زمان بر شدن تمیز سازی بیش از زمان اعلامی در برنامه زمان بندی، مدت			

	• مدت زمان 44 ساعت Rig service مورد بررسی قرار بگیرد			زمان تحمیلی در قالب اضافه کار به کارفرمای محترم اعلام خواهد شد. • مدت زمان Rig Service با احتساب نیم ساعت در روز لحاظ گردیده و قابل تغییر نمی‌باشد.			
3	**Sequence of Operation:** • در یک آیتم روش بستن چاه حتما قید شود.	45	CM	• انجام شد.			
4	**Handover Data:** • فرم های زیر اضافه گردد: • Logging result • BOP test report • Service company report	78	CM	• انجام شد.			
5	**Table of comment:** • جدول کامنت های ارسالی به انتهای برنامه اضافه شود			• انجام شد.			
Fluid							
6	صفحه 53: کدورت سیال برگشتی از چاه قید شده است. استفاده از سیال برگشتی با این مشخصات بر خلاف قرارداد		CM	• سیال ورودی به چاه با مشخصات قید شده در برنامه خواهد بود.			

	است.						
7	در نتیجه ی بررسی ویرایش 00، مقرر گردیده است که برنامه پسماند جداگانه توسط پیمان کار ارسال گردد.		CM	• ارسال خواهد گردید.			